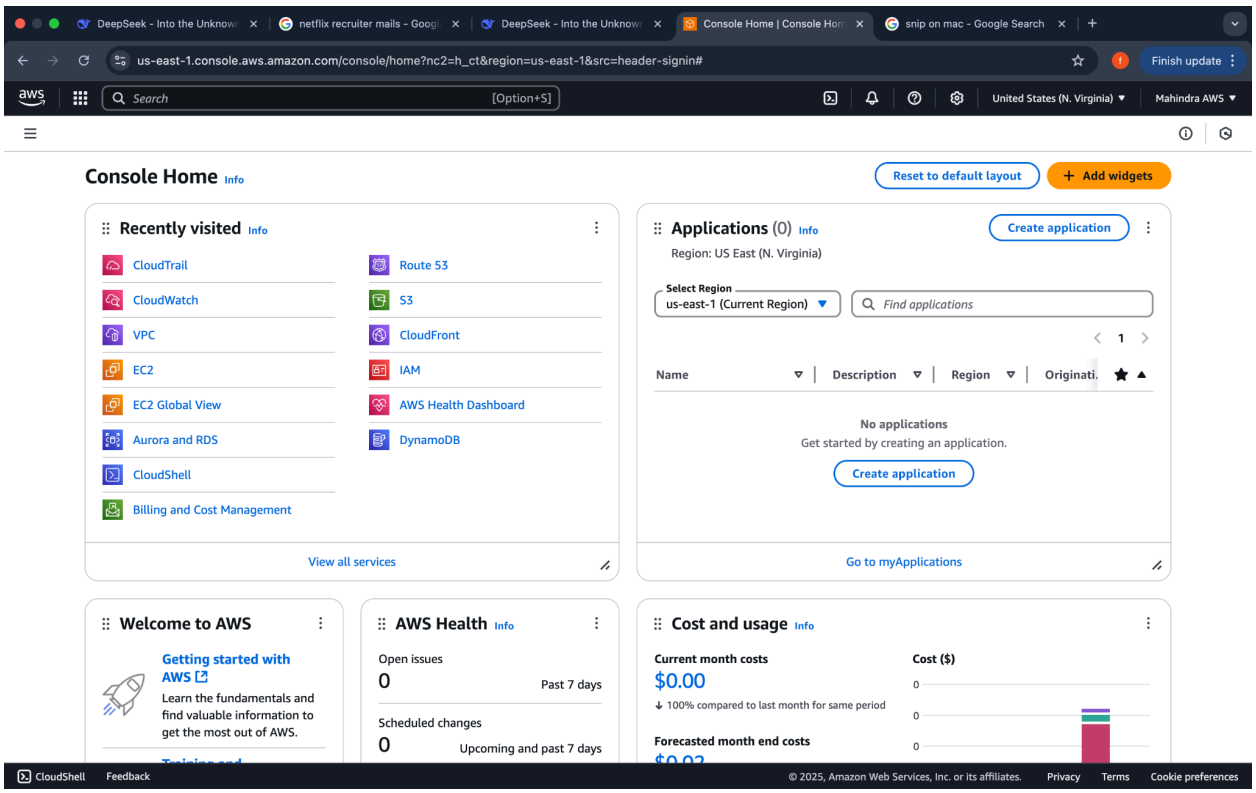
Secure Architectures

Author : Mahindra Guptha Kotha
Mail ID: mkotha@smu.edu

EXERCISE 12.1

Create a Limited Administrative User

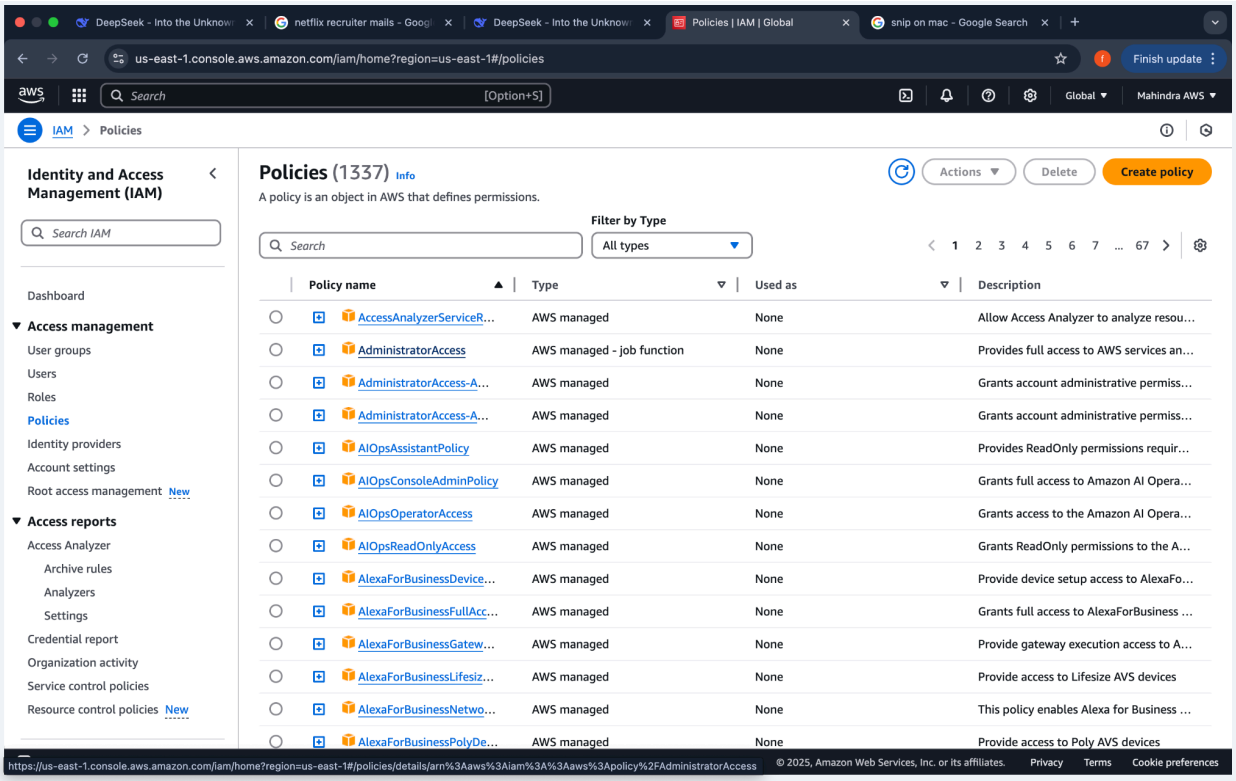
In this exercise, you'll create an IAM user with limited administrative privileges. You'll use policy boundary permissions to allow the user to perform only actions in the EC2 service.

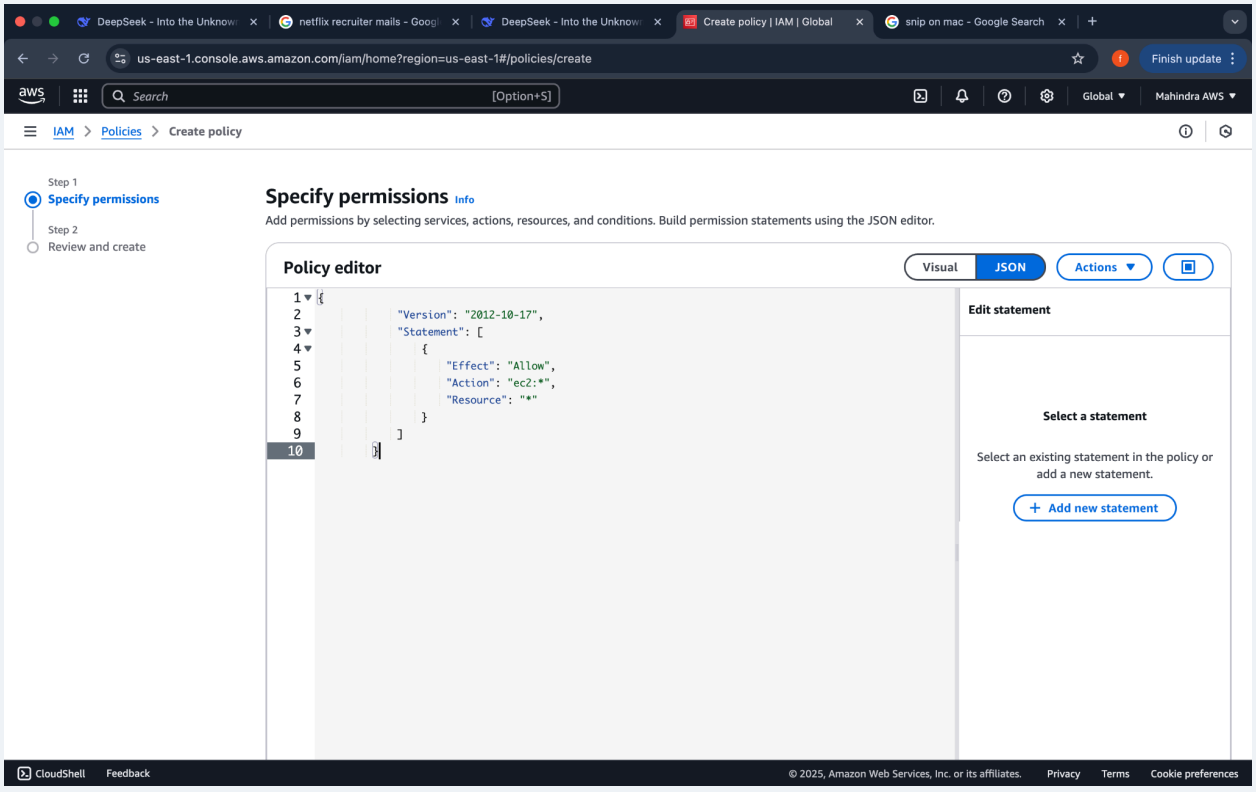


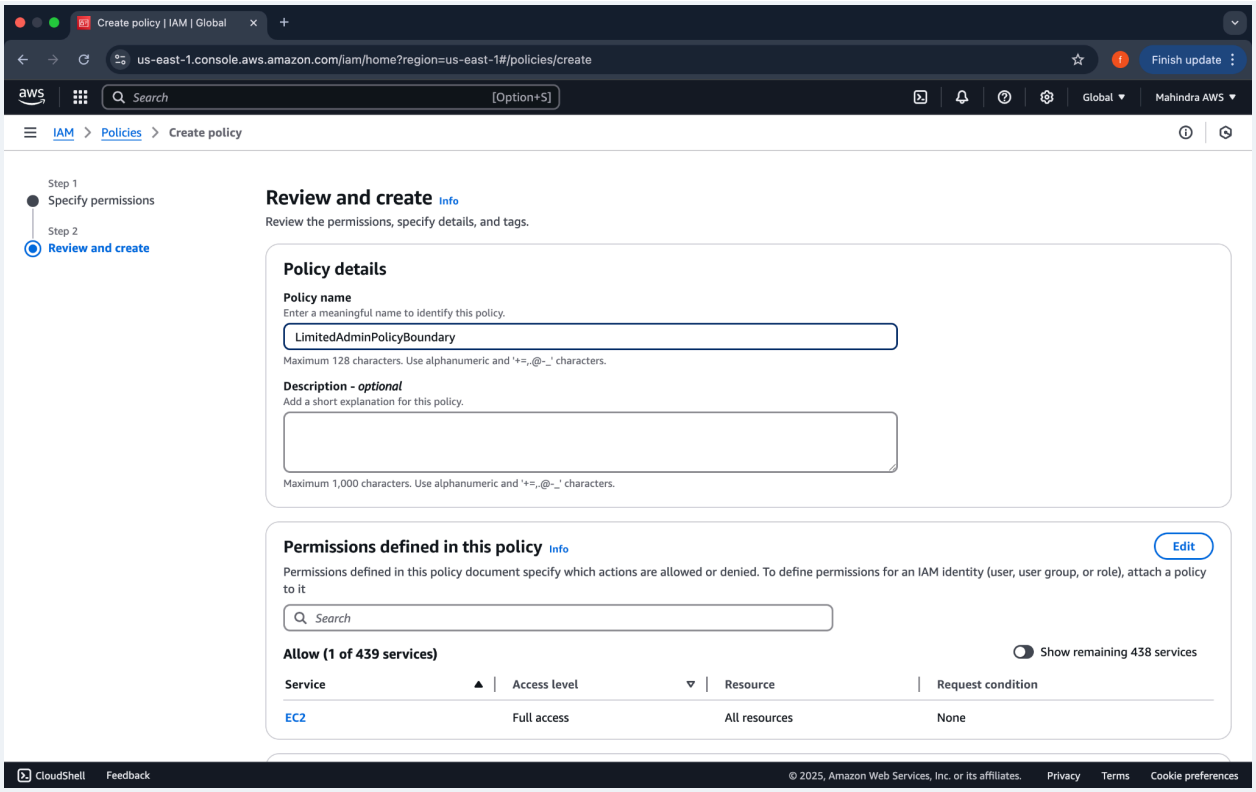
1. Create a customer-managed policy called `LimitedAdminPolicyBoundary`. Populate the policy document with the following content:

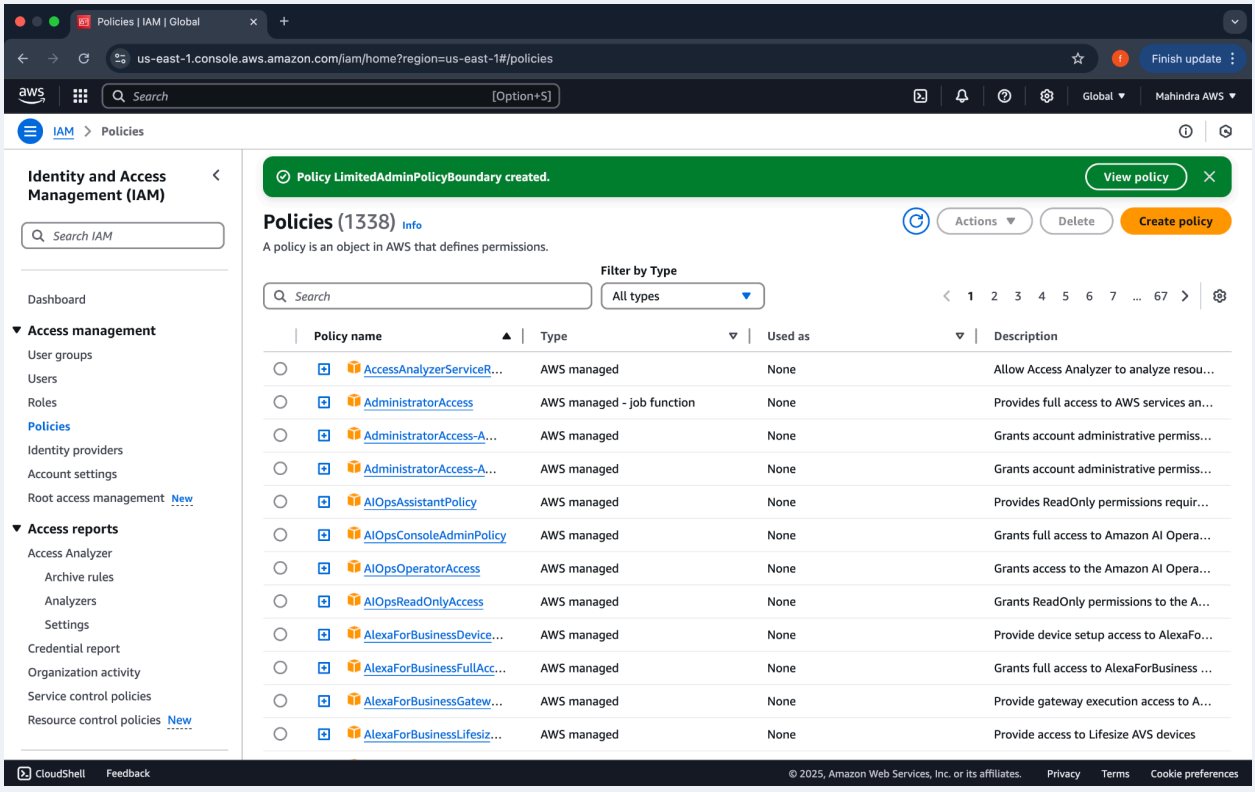
```
{
```

```
            "Version": "2012-10-17",
            "Statement": [
                {
                    "Effect": "Allow",
                    "Action": "ec2:*",
                    "Resource": "*"
                }
            ]
        }
```



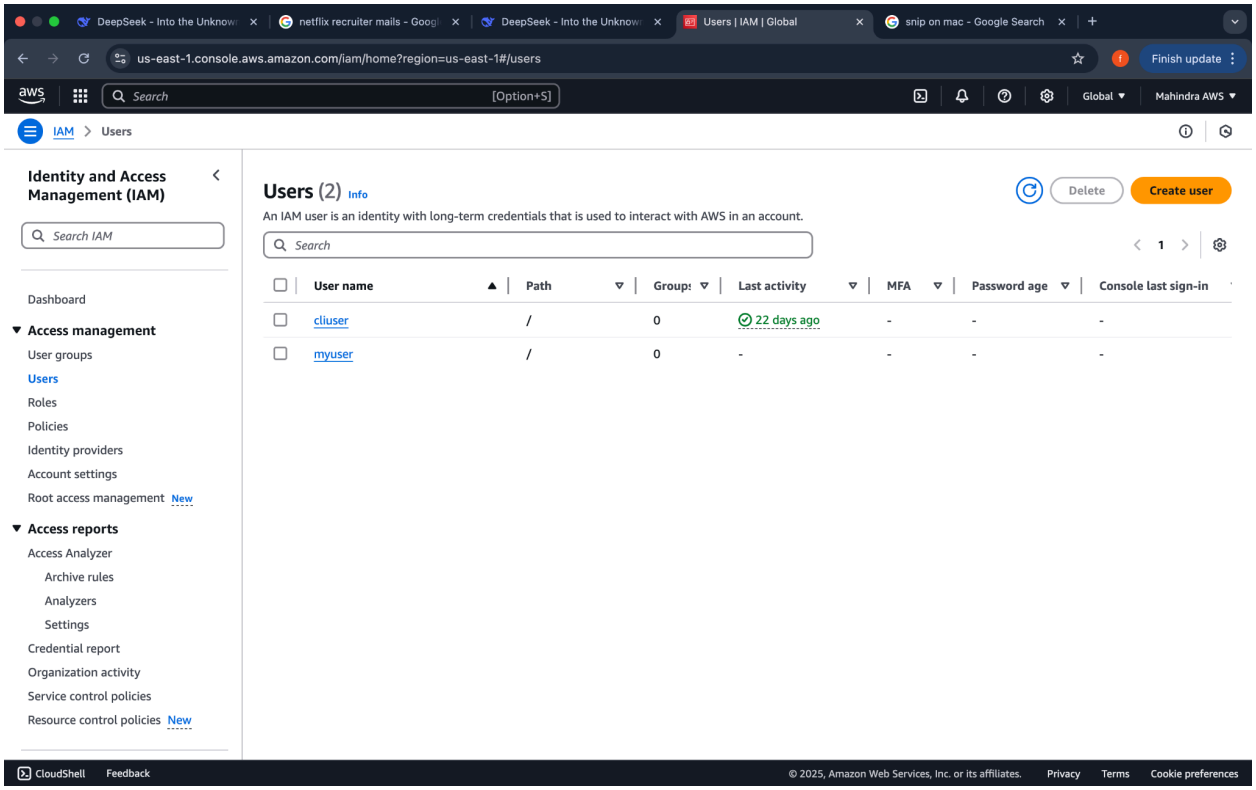

DeepSeek - Into the Unknow
netflix recruiter mails - Goog
DeepSeek - Into the Unknow
Create policy | IAM | Global
snip on mac - Google Search
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/policies/create
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Policies > Create policy
Step 1
Specify permissions
Step 2
Review and create
Specify permissions Info
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": "ec2:*",
"Resource": "*"
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
Add new statement
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

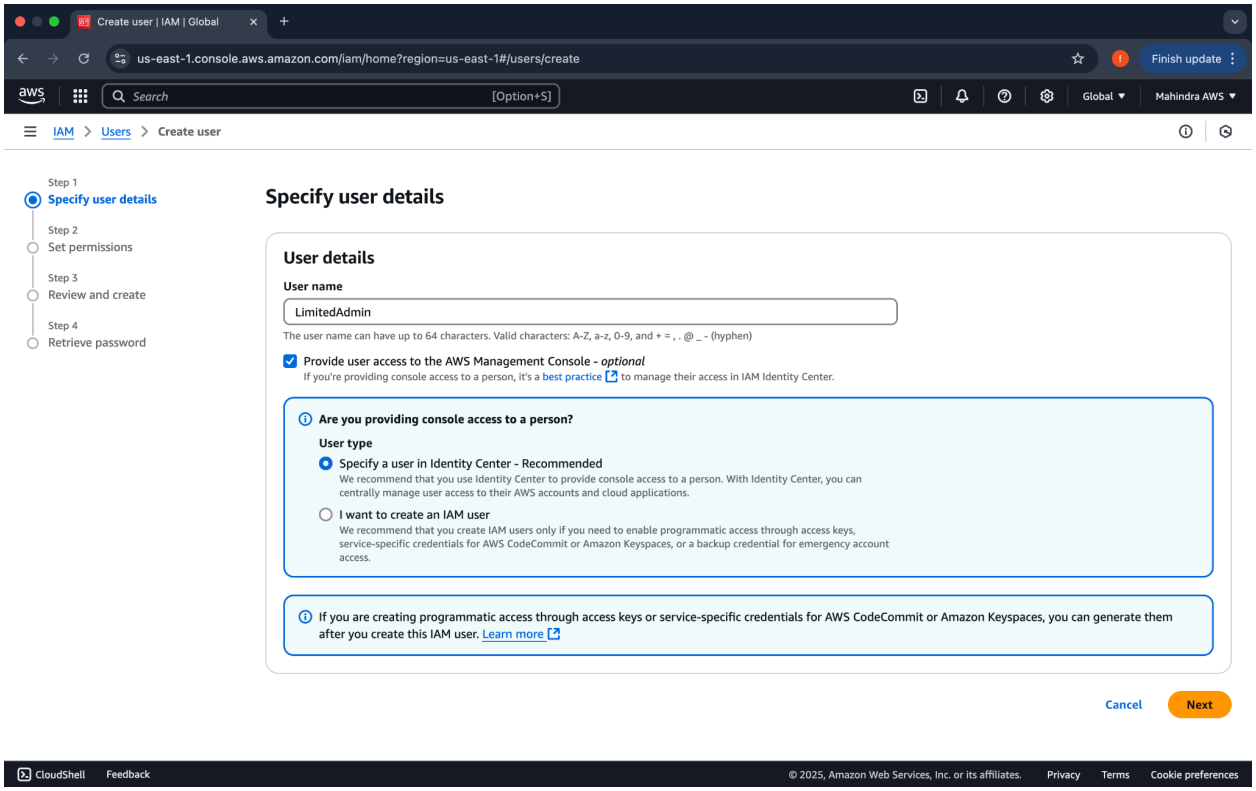

Create policy | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/policies/create
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Policies > Create policy
Step 1
Specify permissions
Step 2
Review and create
Review and create Info
Review the permissions, specify details, and tags.
Policy details
Policy name
Enter a meaningful name to identify this policy.
LimitedAdminPolicyBoundary
Maximum 128 characters. Use alphanumeric and '+=,.@-_' characters.
Description - optional
Add a short explanation for this policy.
Maximum 1,000 characters. Use alphanumeric and '+=,.@-_' characters.
Permissions defined in this policy Info
Edit
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
Search
Allow (1 of 439 services)
Show remaining 438 services
Service
Access level
Resource
Request condition
EC2
Full access
All resources
None
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

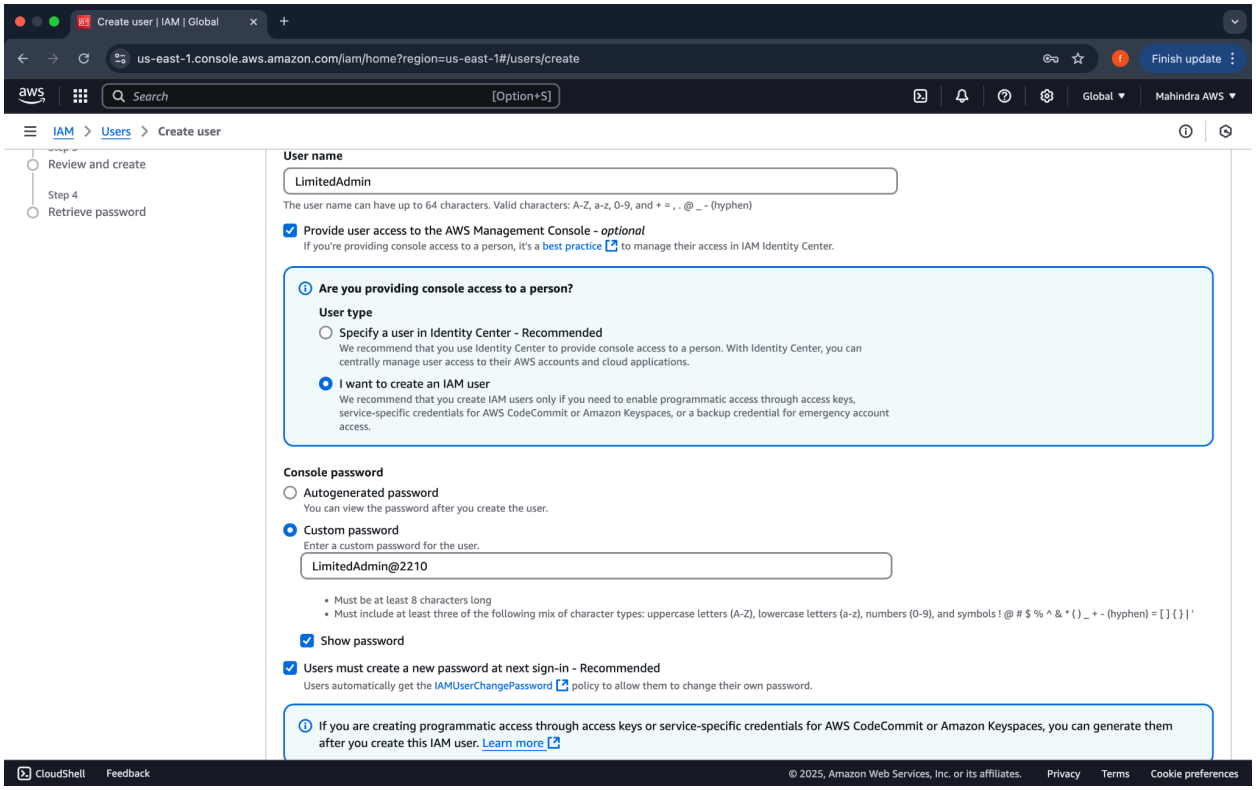


Create an IAM user with the name of your choice, such as LimitedAdmin.

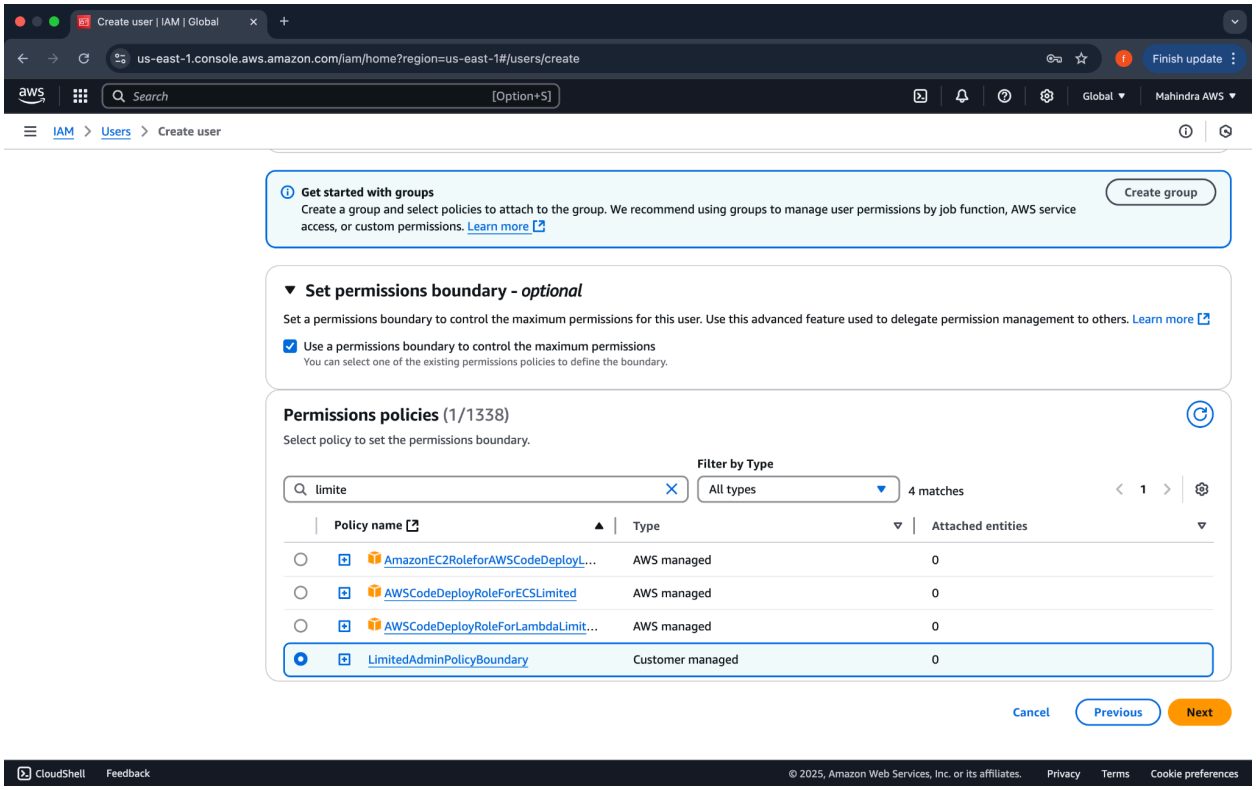
1. In the IAM Dashboard, click the Users option and then click the username you created.


DeepSeek - Into the Unknow
netflix recruiter mails - Googl
DeepSeek - Into the Unknow
Users | IAM | Global
snip on mac - Google Search
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
Access reports
Access Analyzer
Archive rules
Analyzers
Settings
Credential report
Organization activity
Service control policies
Resource control policies New
Users (2) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete
Create user
Search
1
User name
Path
Groups
Last activity
MFA
Password age
Console last sign-in
cliuser
/
0
22 days ago
-
-
-
myuser
/
0
-
-
-
-
CloudShell
Feedback
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Terms
Cookie preferences


Create user | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Users > Create user
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Specify user details
User details
User name
LimitedAdmin
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)
Provide user access to the AWS Management Console - optional
If you're providing console access to a person, it's a best practice to manage their access in IAM Identity Center.
Are you providing console access to a person?
User type
Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to their AWS accounts and cloud applications.
I want to create an IAM user
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.
If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more
Cancel
Next
CloudShell
Feedback
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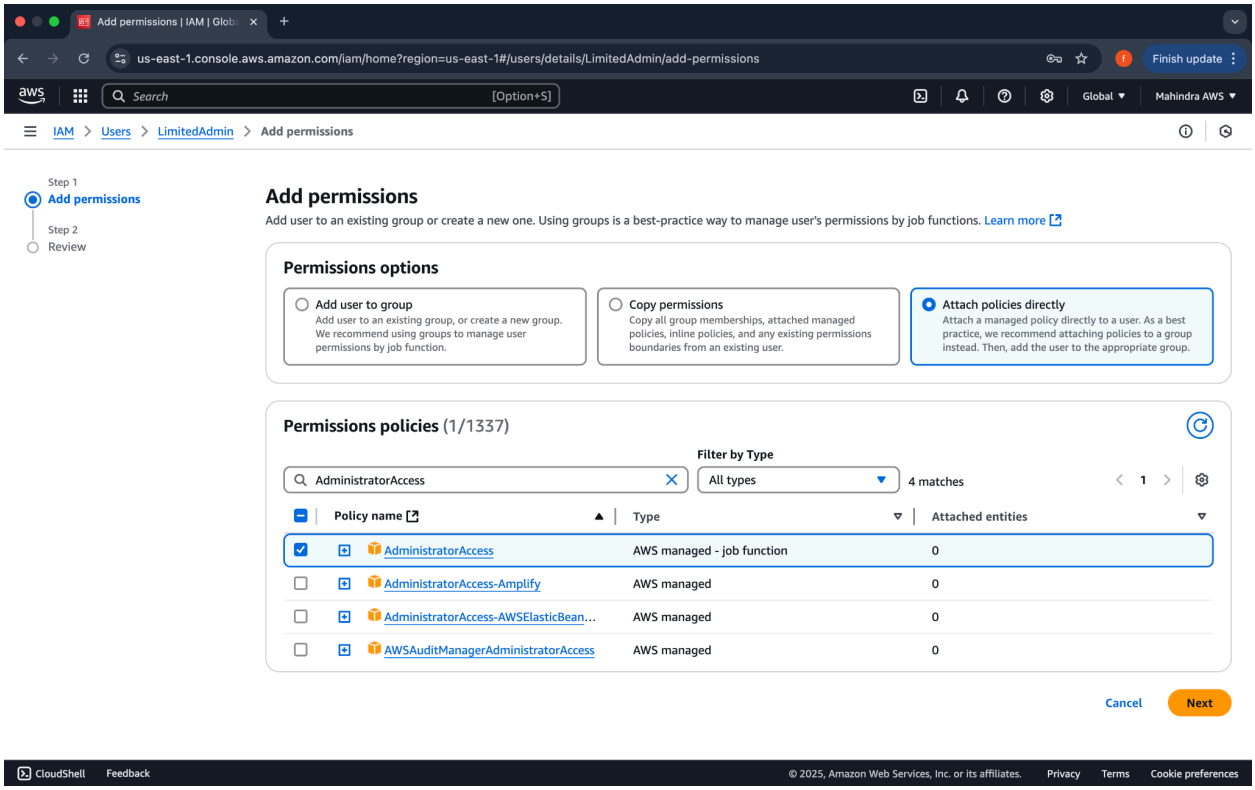


2. Under Permissions Boundary, click the Set Boundary button.
3. Select the LimitedAdminPolicyBoundary option and click the Set Boundary button.

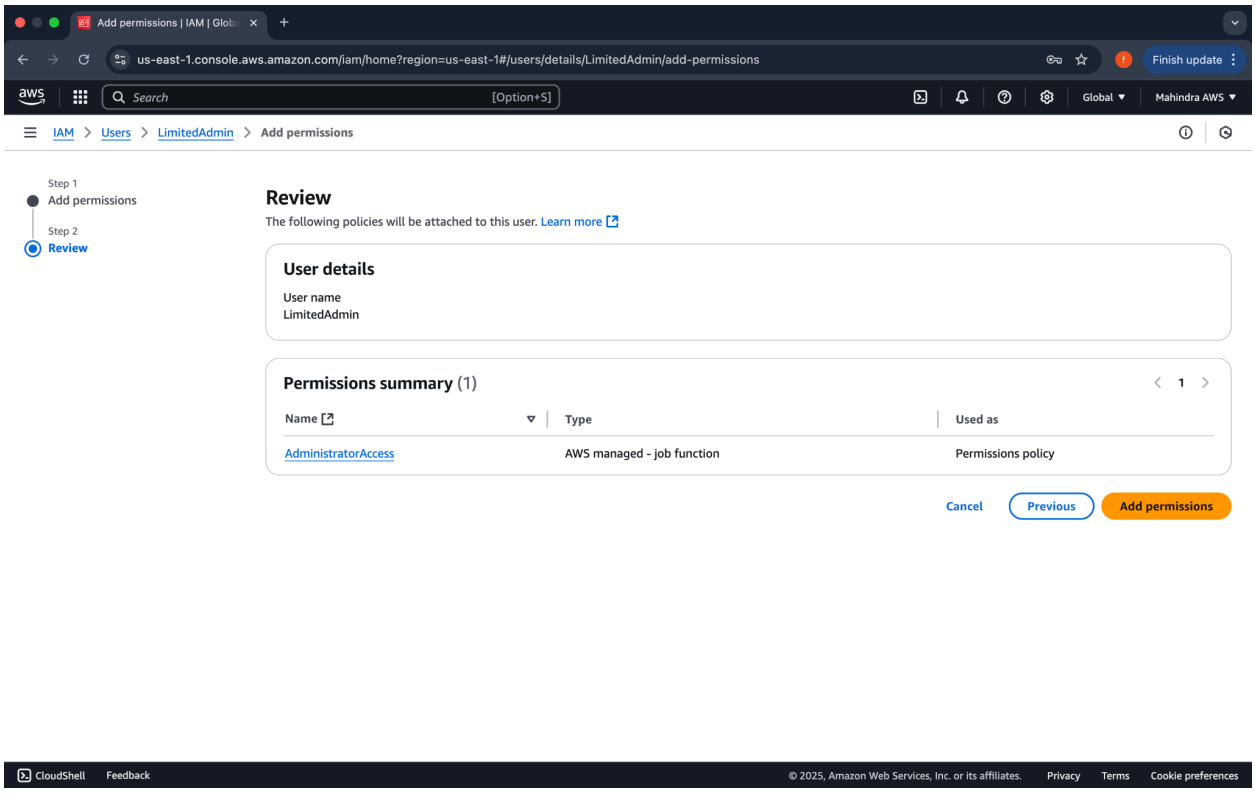


4. Under Permissions Policy, click Add Permissions.
5. Click the Attach Existing Policies Directly button.
6. Select the AdministratorAccess policy and click the Next: Review button.

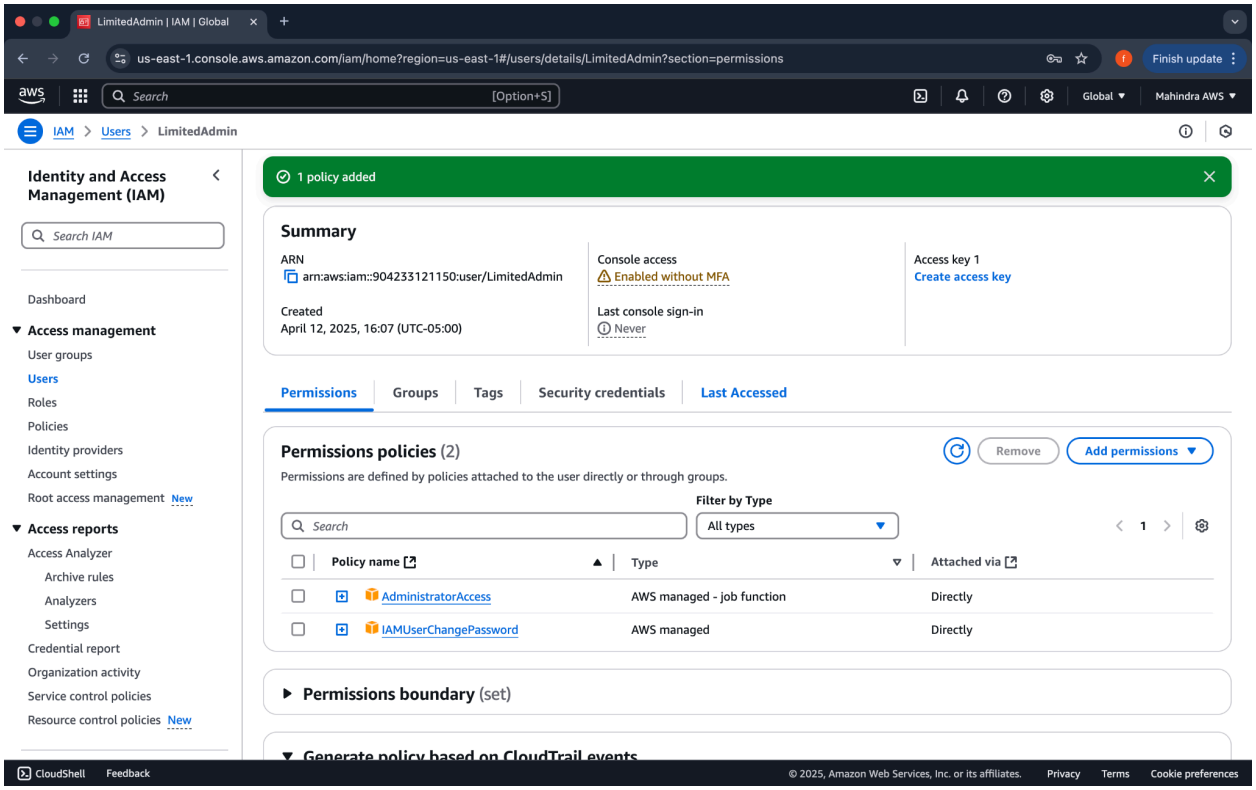
7.



8.



9. Click the Add Permissions button.

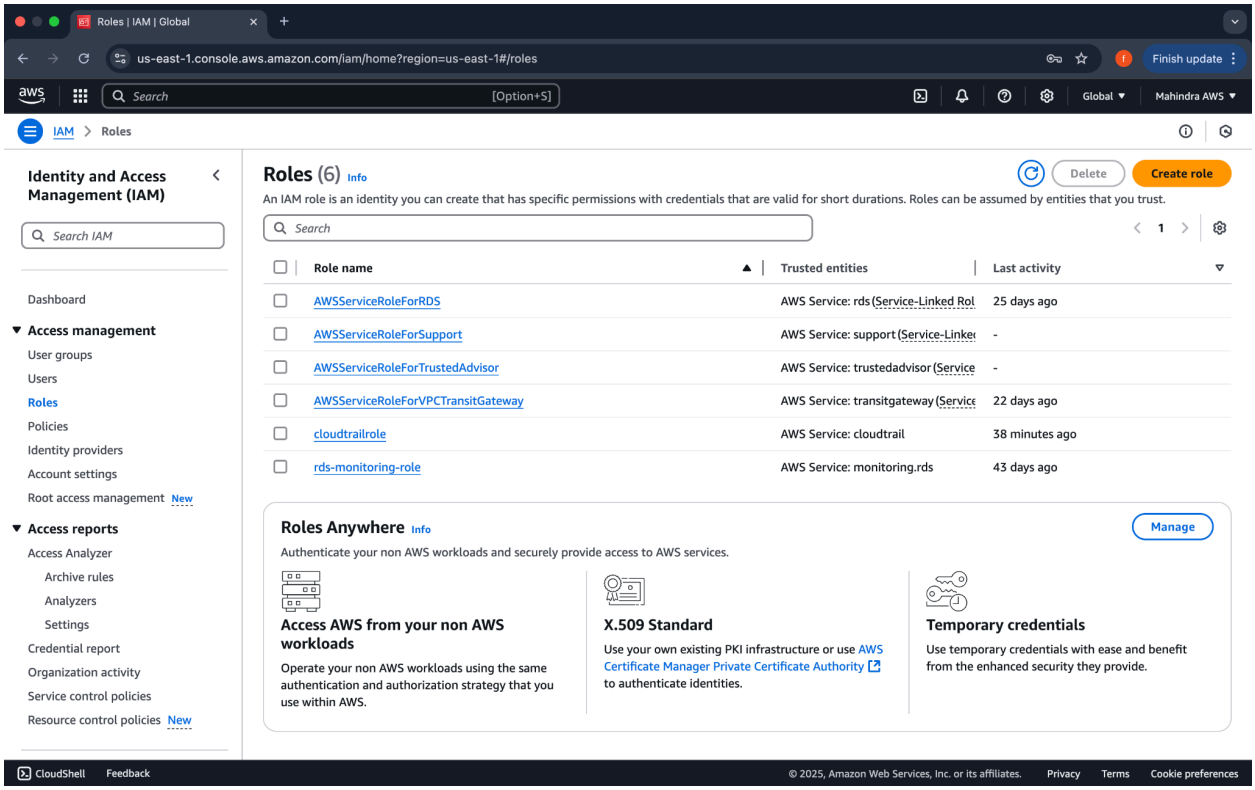


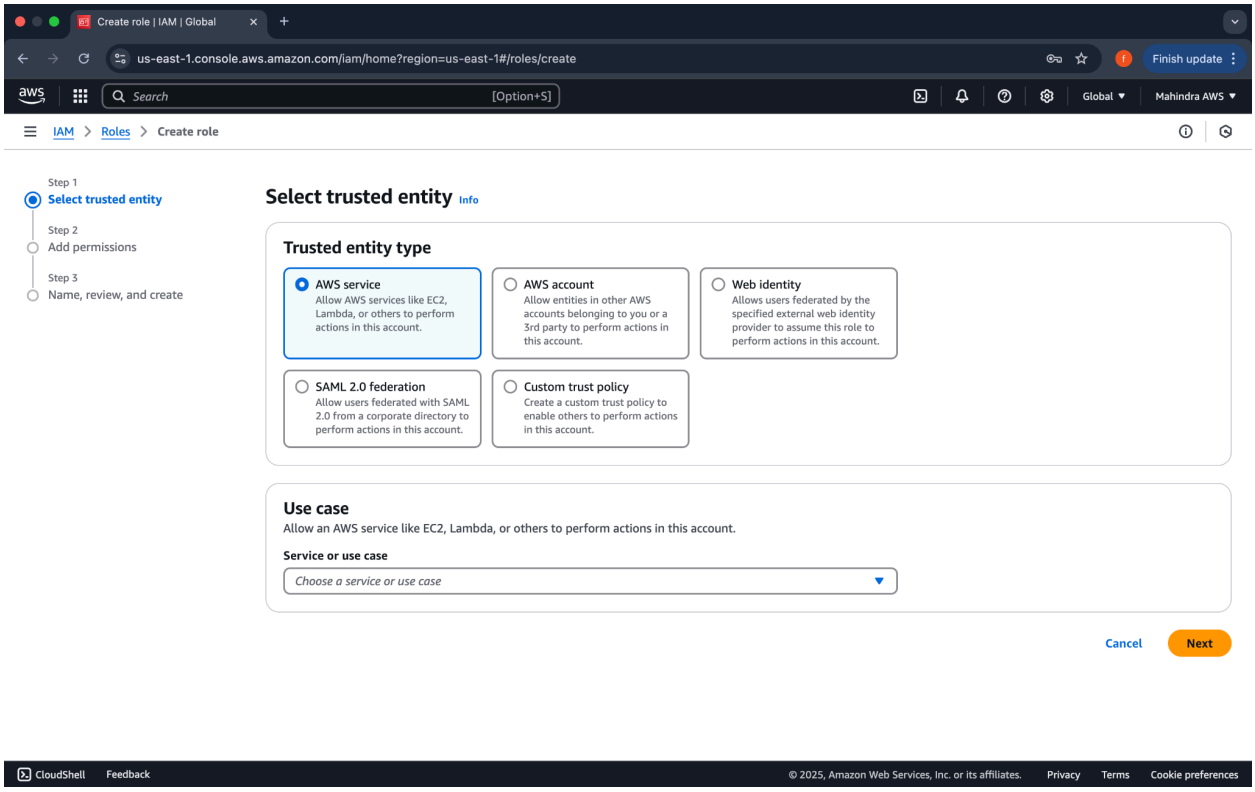
EXERCISE 12.2

Create and Assume a Role as an IAM User

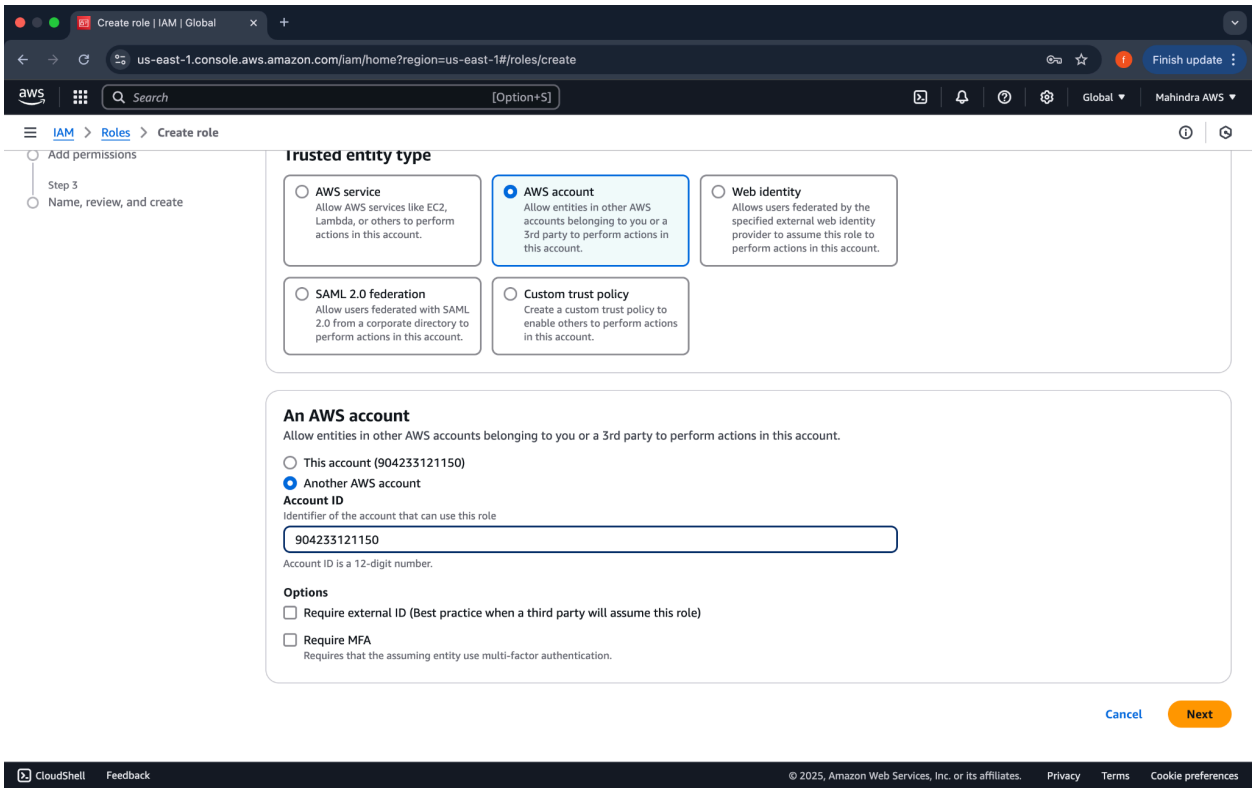
In this exercise, you'll create a role and assume it.

1. While logged in as an administrative IAM user, click Roles on the menu on the left side of the IAM Dashboard screen and then click the Create Role button.

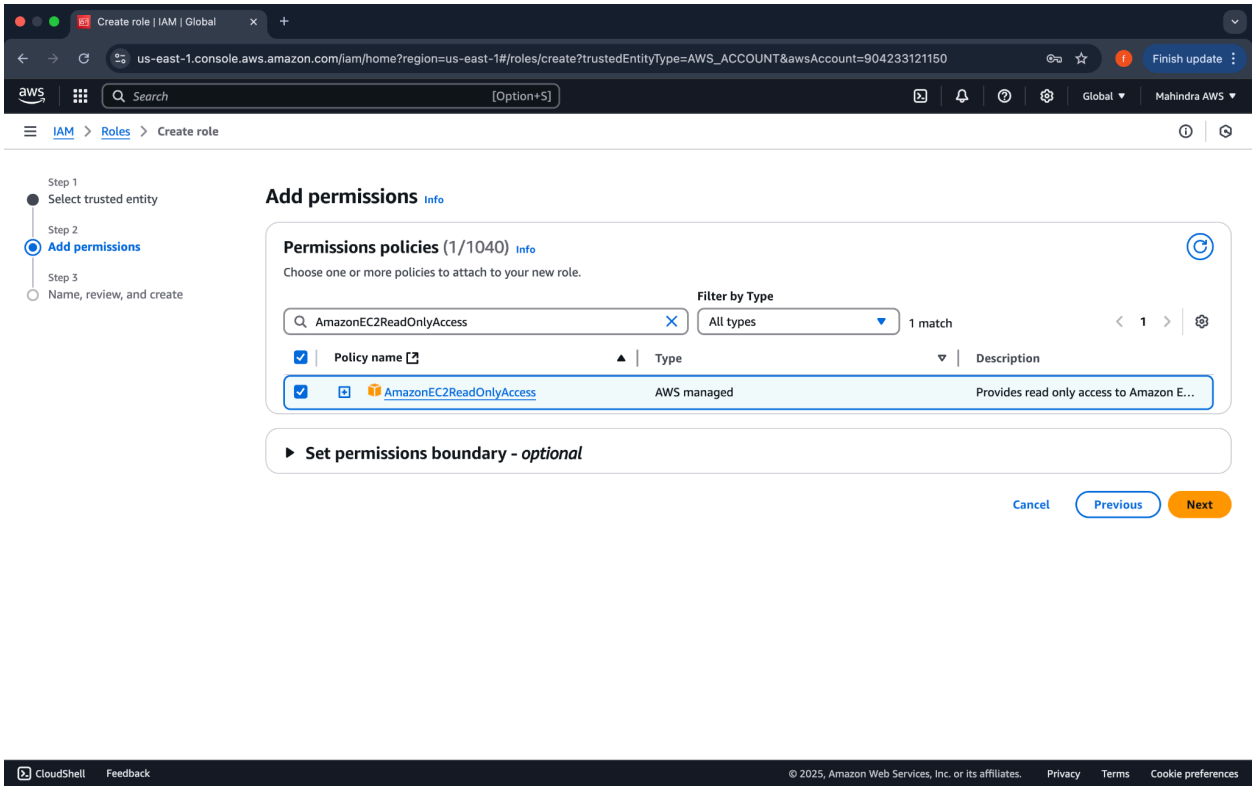

Roles | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management New
Access reports
Access Analyzer
Archive rules
Analyzers
Settings
Credential report
Organization activity
Service control policies
Resource control policies New
Roles (6) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
Search
1
Role name
Trusted entities
Last activity
AWSServiceRoleForRDS
AWS Service: rds (Service-Linked Rol
25 days ago
AWSServiceRoleForSupport
AWS Service: support (Service-Linke
-
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Service
-
AWSServiceRoleForVPCTransitGateway
AWS Service: transitgateway (Servic
22 days ago
cloudtrailrole
AWS Service: cloudtrail
38 minutes ago
rds-monitoring-role
AWS Service: monitoring.rds
43 days ago
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.
Access AWS from your non AWS workloads
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
X.509 Standard
Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.
Temporary credentials
Use temporary credentials with ease and benefit from the enhanced security they provide.
CloudShell
Feedback
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Terms
Cookie preferences


Create role | IAM | Global
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create
Finish update
Search
[Option+S]
Global
Mahindra AWS
IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
Choose a service or use case
Cancel
Next
CloudShell
Feedback
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Terms
Cookie preferences

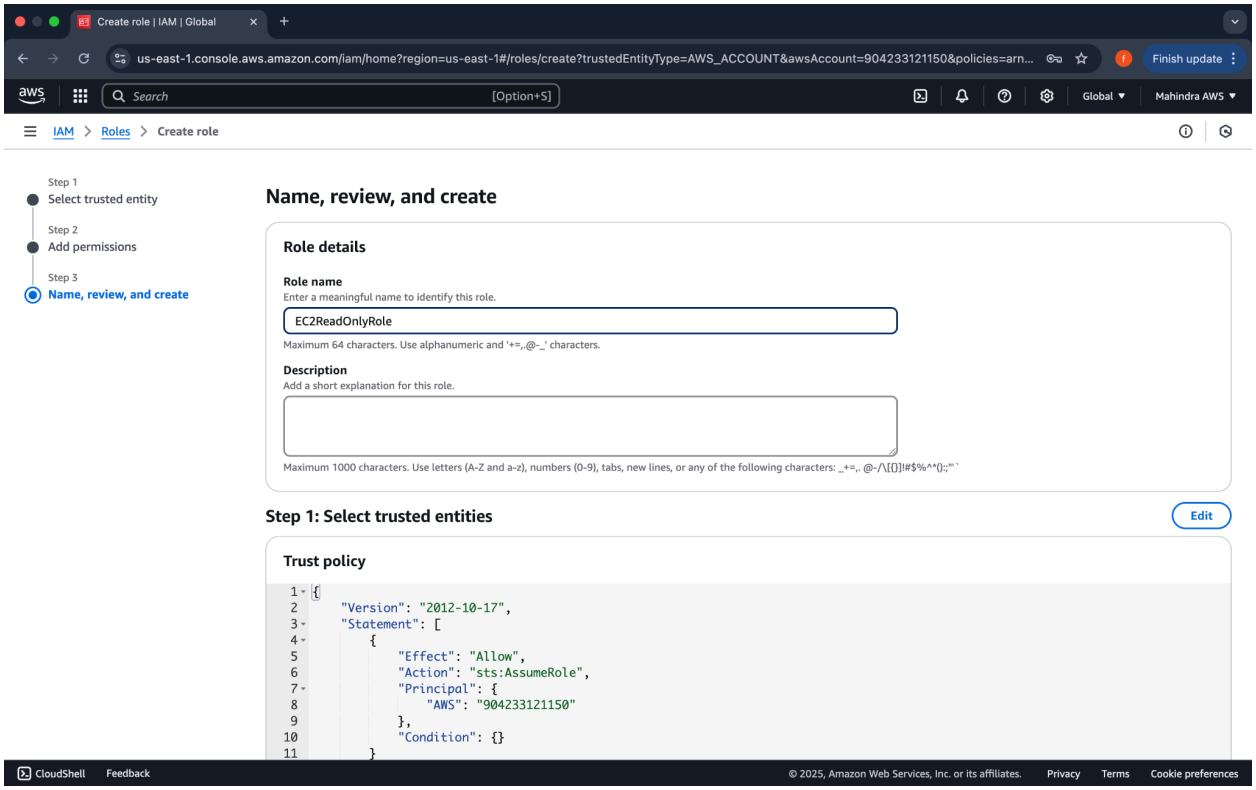
2. Under Select Type Of Trusted Entity, click the Another AWS Account button.
3. Under Account ID, enter your AWS account number. Click the Next: Permissions button.



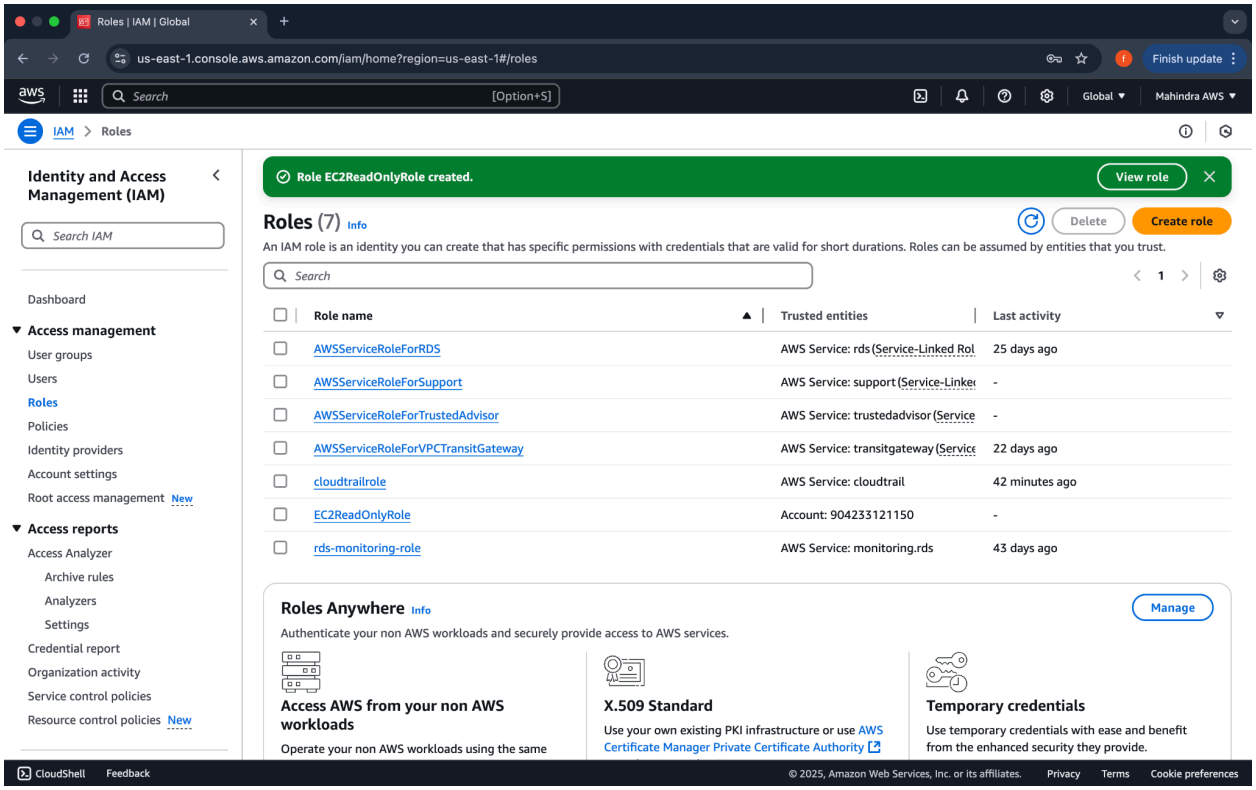
4. Select the AmazonEC2ReadOnlyAccess AWS managed policy and click the Next: Tags button.



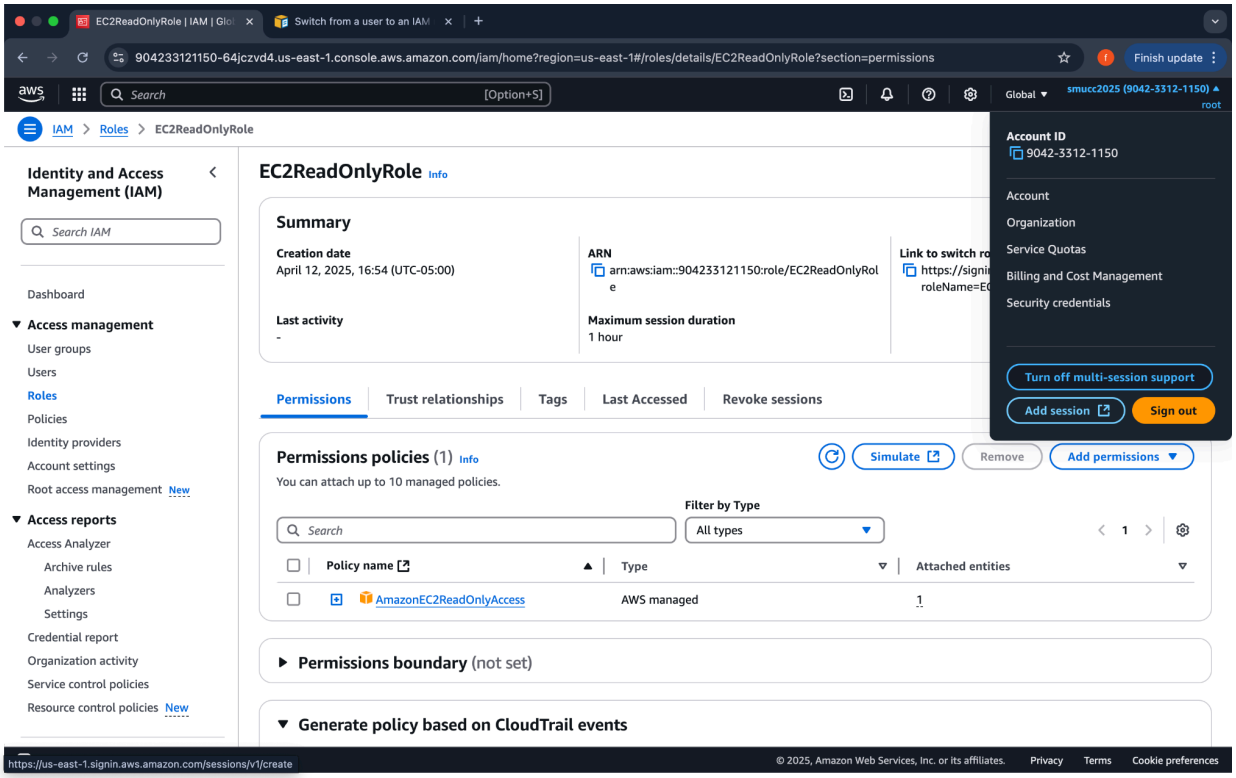
5. Click the Next: Review button.
6. Enter a name for the role, such as **EC2ReadOnlyRole**.



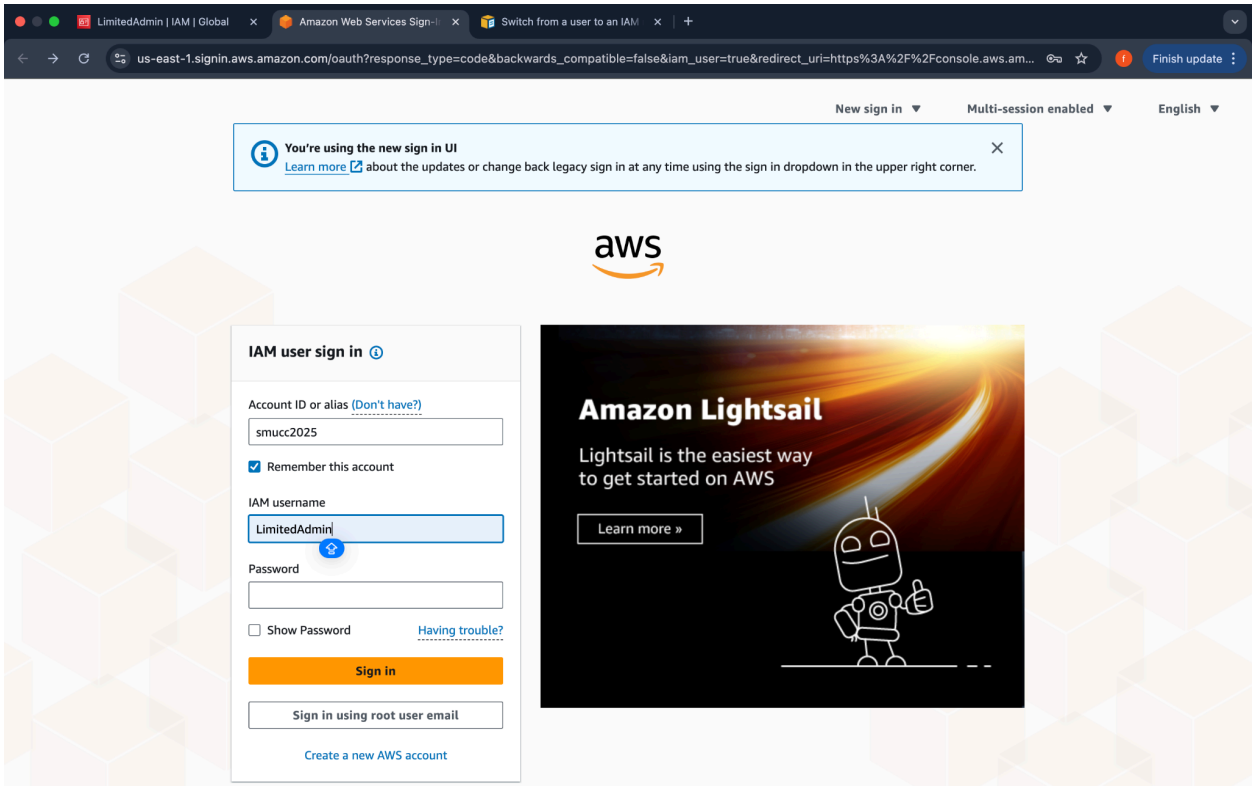
7. Click the Create Role button.



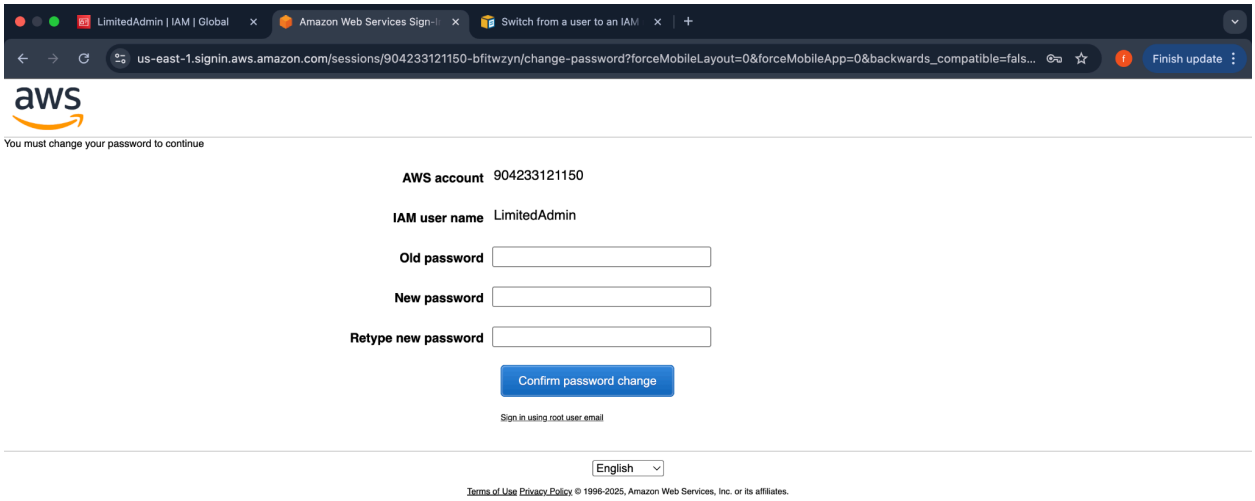
8. On the navigation bar at the top of the AWS Management Console, click your IAM account name. A drop-down menu will appear.
9. Click the Switch Role link in the drop-down menu, as shown here



10. Click the Switch Role button.
11. Enter your 12-digit AWS account number or alias and the name of the role you created in step 6. The following shows an example of a Switch Role screen with the account and role specified.



12. Click the Switch Role button. The name of the role followed will appear in the top navigation bar. You will now be operating under the role you created.
13. Try to launch an EC2 instance. It will fail because the assumed role doesn't have the `RunInstances` permission.



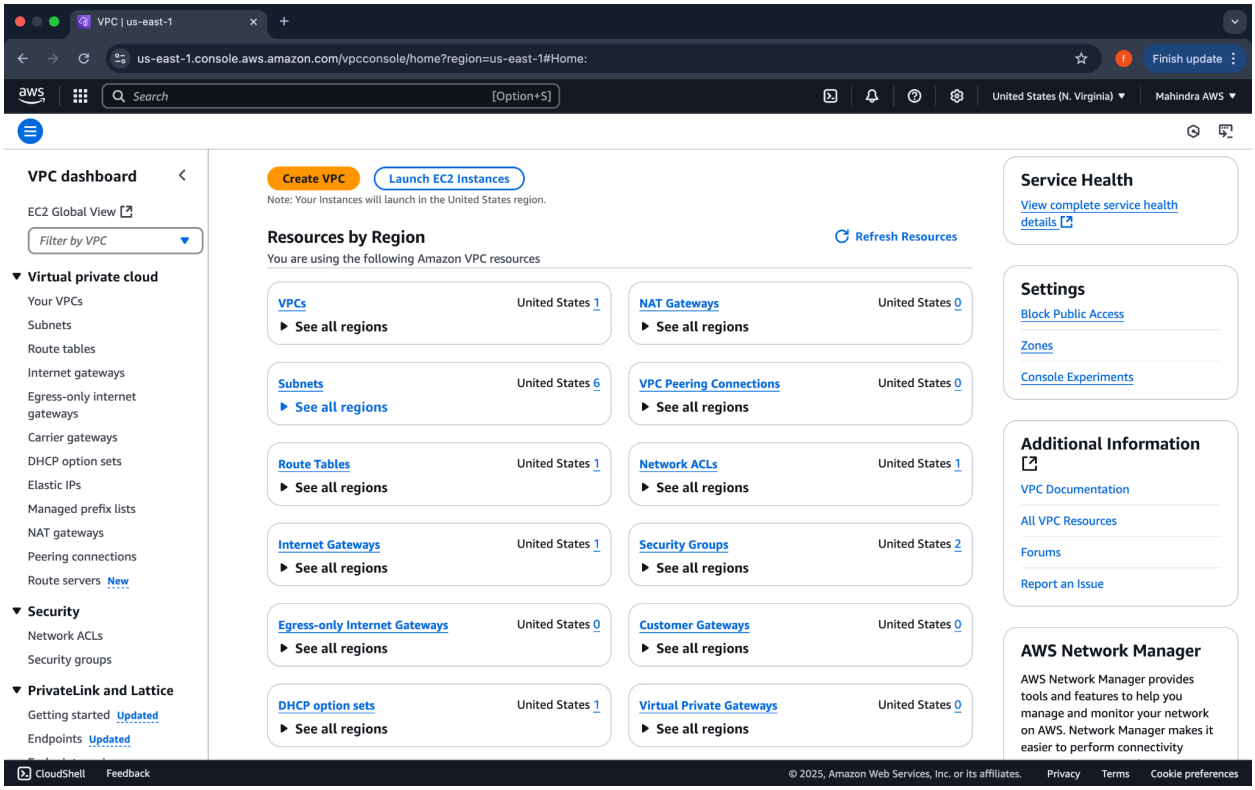
14. To switch back to your IAM user, click the upper-right menu bar where the name of the role followed by your AWS account number is. Then click the link that reads Back To *[your IAM username]*.

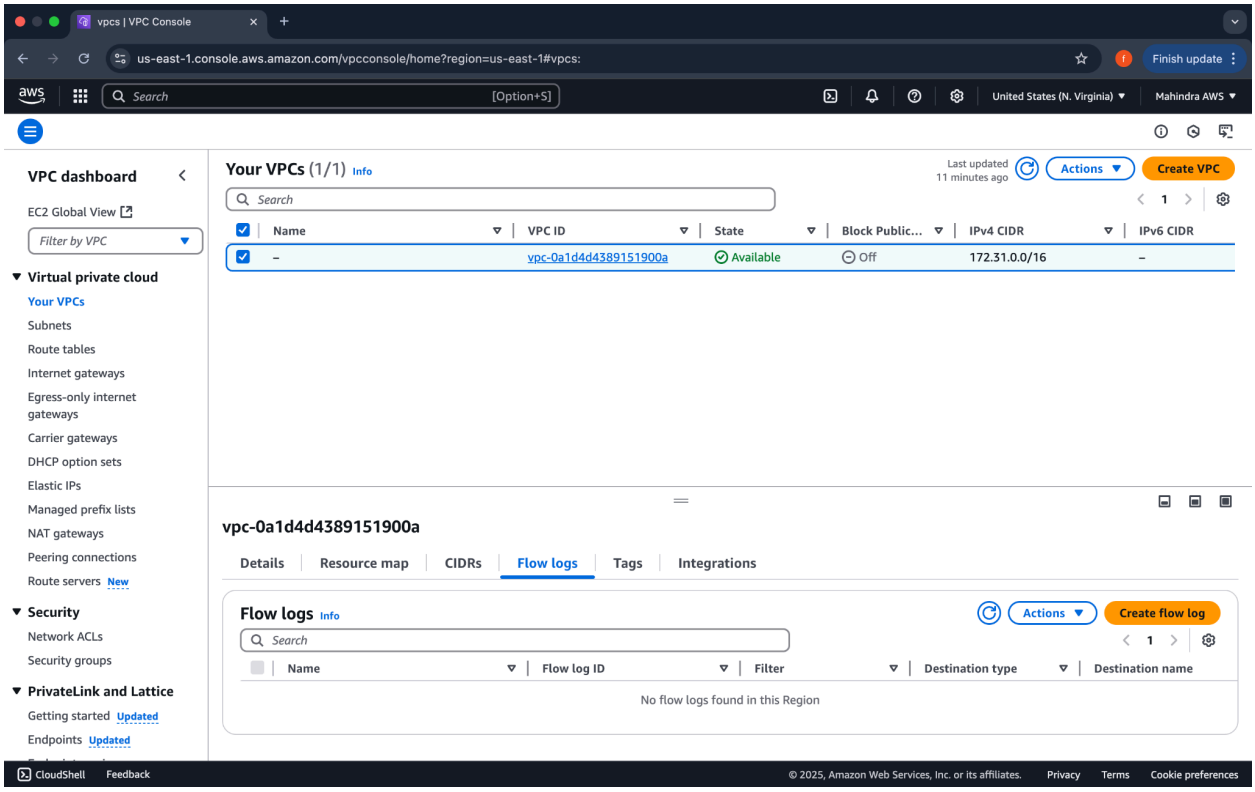
EXERCISE 12.3

Configure VPC Flow Logging

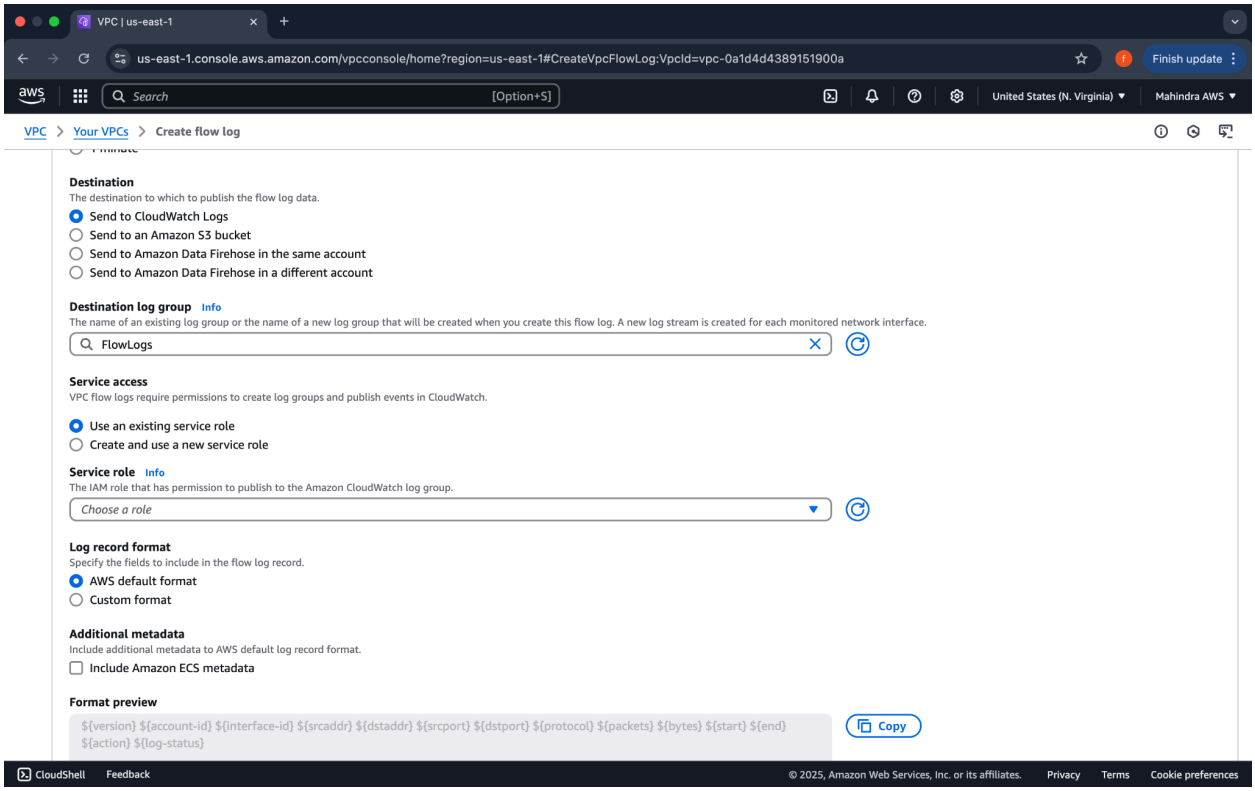
In this exercise, you'll configure VPC flow logging to deliver flow logs to a CloudWatch Logs log group.

1. In the VPC Dashboard, select a VPC you want to log traffic for. Click the Flow Logs tab.

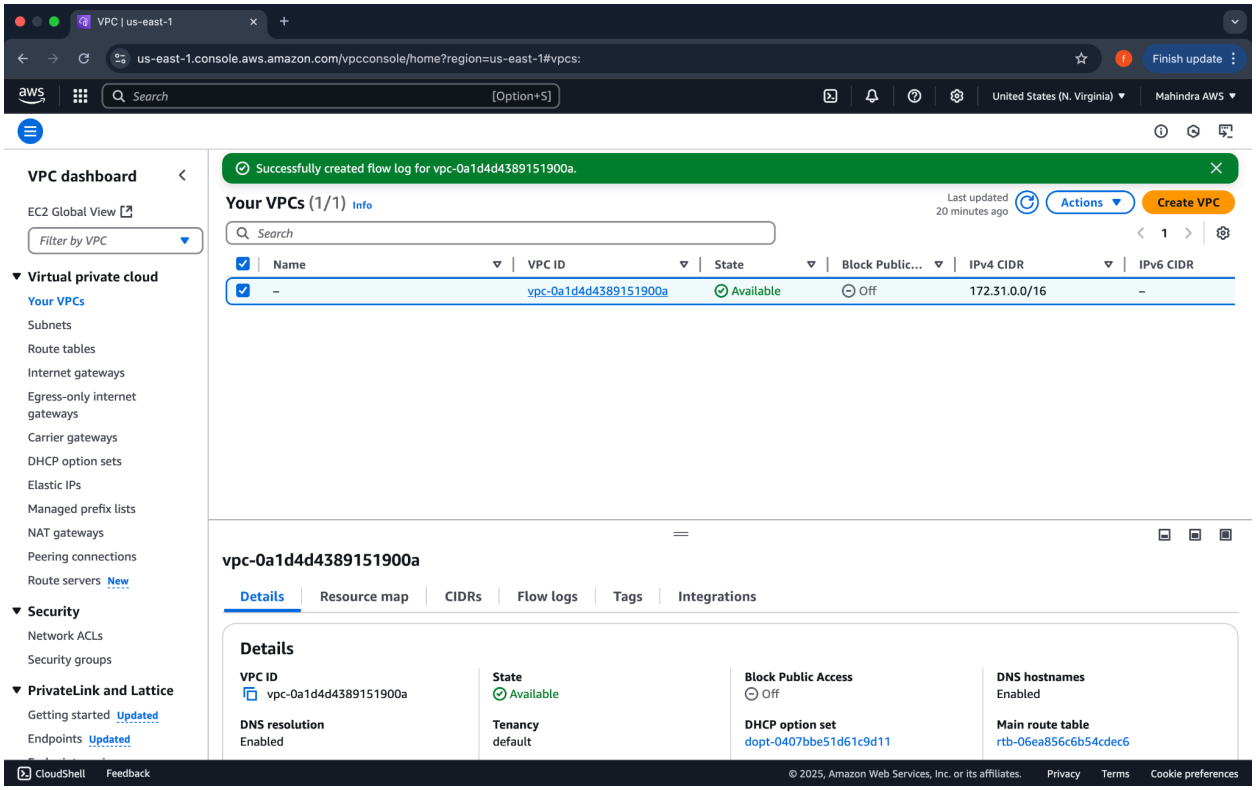

VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Home:
Finish update
Search [Option+S]
United States (N. Virginia)
Mahindra AWS
VPC dashboard
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers New
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started Updated
Endpoints Updated
Create VPC
Launch EC2 Instances
Note: Your Instances will launch in the United States region.
Resources by Region
Refresh Resources
You are using the following Amazon VPC resources
VPCs United States 1
See all regions
NAT Gateways United States 0
See all regions
Subnets United States 6
See all regions
VPC Peering Connections United States 0
See all regions
Route Tables United States 1
See all regions
Network ACLs United States 1
See all regions
Internet Gateways United States 1
See all regions
Security Groups United States 2
See all regions
Egress-only Internet Gateways United States 0
See all regions
Customer Gateways United States 0
See all regions
DHCP option sets United States 1
See all regions
Virtual Private Gateways United States 0
See all regions
Service Health
View complete service health details
Settings
Block Public Access
Zones
Console Experiments
Additional Information
VPC Documentation
All VPC Resources
Forums
Report an Issue
AWS Network Manager
AWS Network Manager provides tools and features to help you manage and monitor your network on AWS. Network Manager makes it easier to perform connectivity
CloudShell
Feedback
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vpcs | VPC Console
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs:
Finish update
Search [Option+S]
United States (N. Virginia)
Mahindra AWS
VPC dashboard
EC2 Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers New
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started Updated
Endpoints Updated
Your VPCs (1/1) Info
Last updated 11 minutes ago
Actions
Create VPC
Search
Name
VPC ID
State
Block Public...
IPv4 CIDR
IPv6 CIDR
–
vpc-0a1d4d4389151900a
Available
Off
172.31.0.0/16
–
vpc-0a1d4d4389151900a
Details
Resource map
CIDRs
Flow logs
Tags
Integrations
Flow logs Info
Actions
Create flow log
Search
Name
Flow log ID
Filter
Destination type
Destination name
No flow logs found in this Region
CloudShell
Feedback
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2. Click the Create Flow Log button.
3. In the Destination Log Group field, enter a name of your choice, such as **FlowLogs**.



4. You'll need to specify a role, which AWS can create for you. Click the Set Up Permissions link.
5. A new browser tab will open. Click the Allow button. AWS will create a role called FlowlogsRole.
6. Return to the previous tab with the Create Flow Log Wizard. Select the FlowlogsRole.
7. Click the Create button.

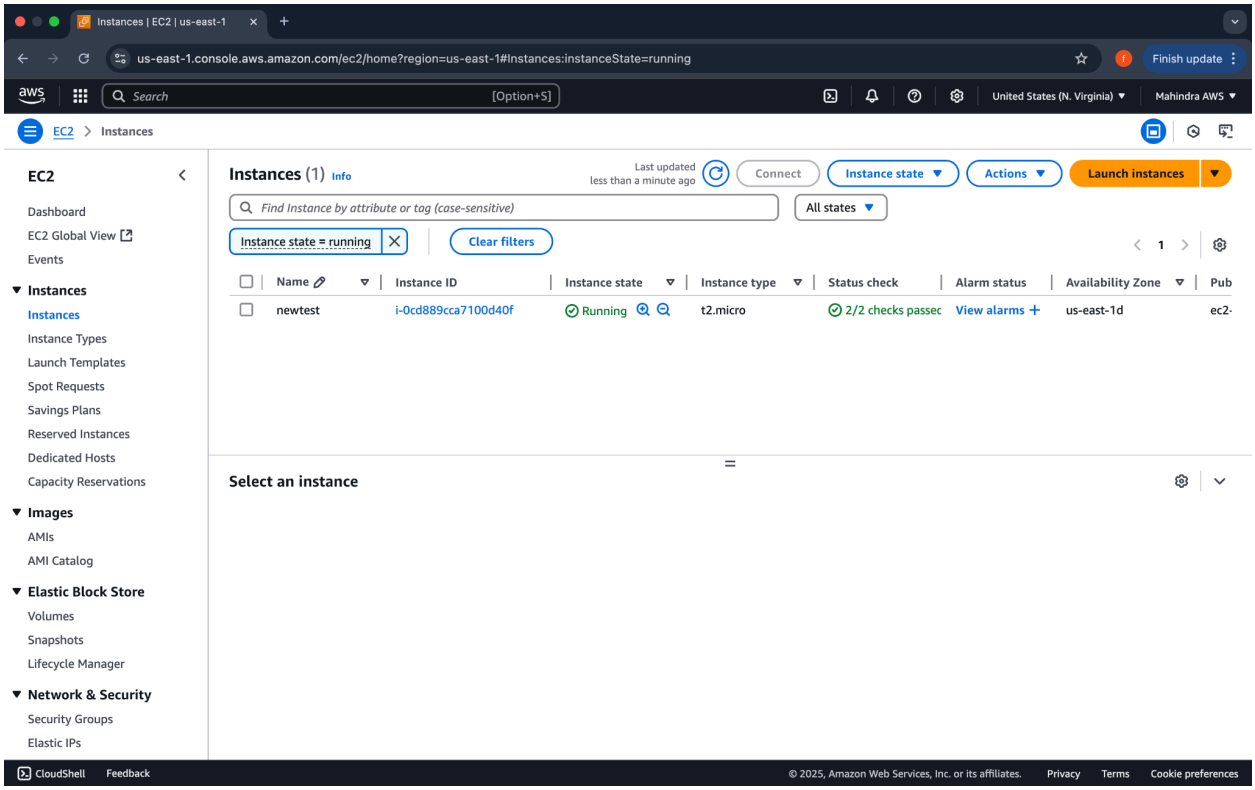


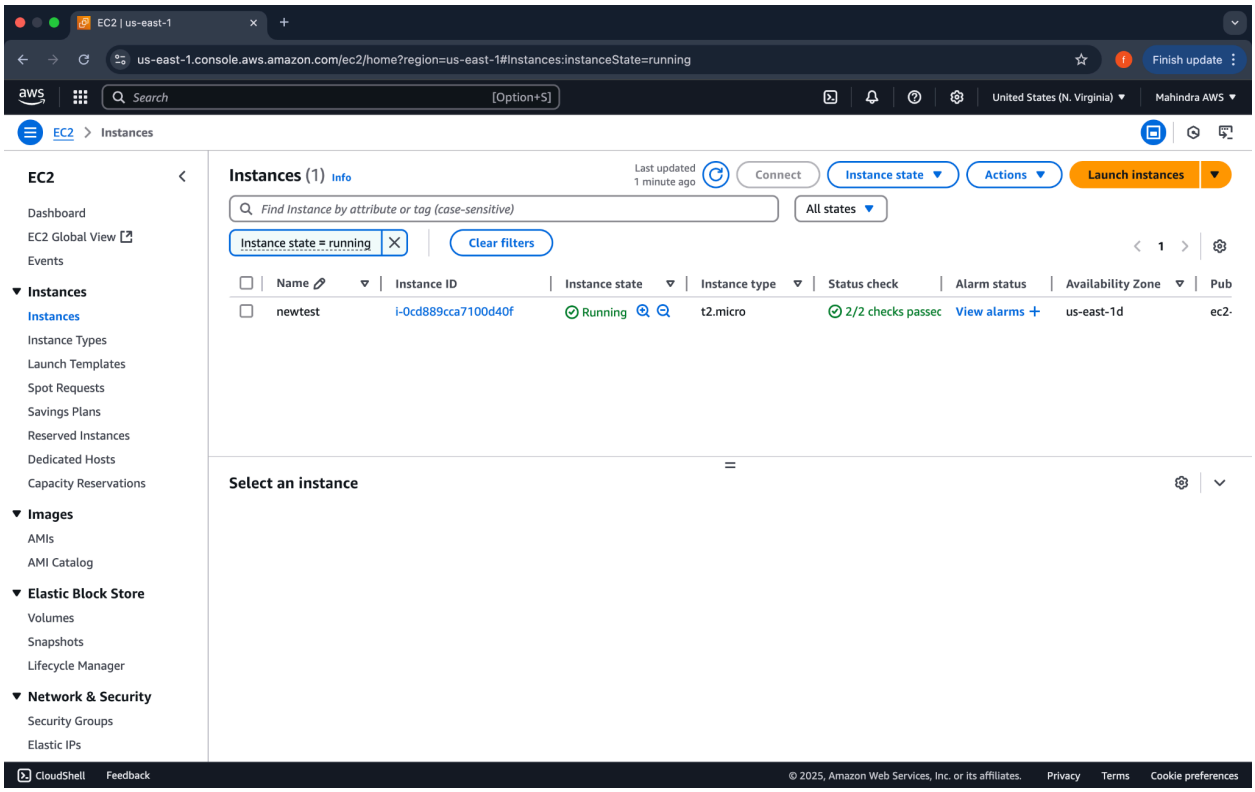
EXERCISE 12.4

Encrypt an EBS Volume

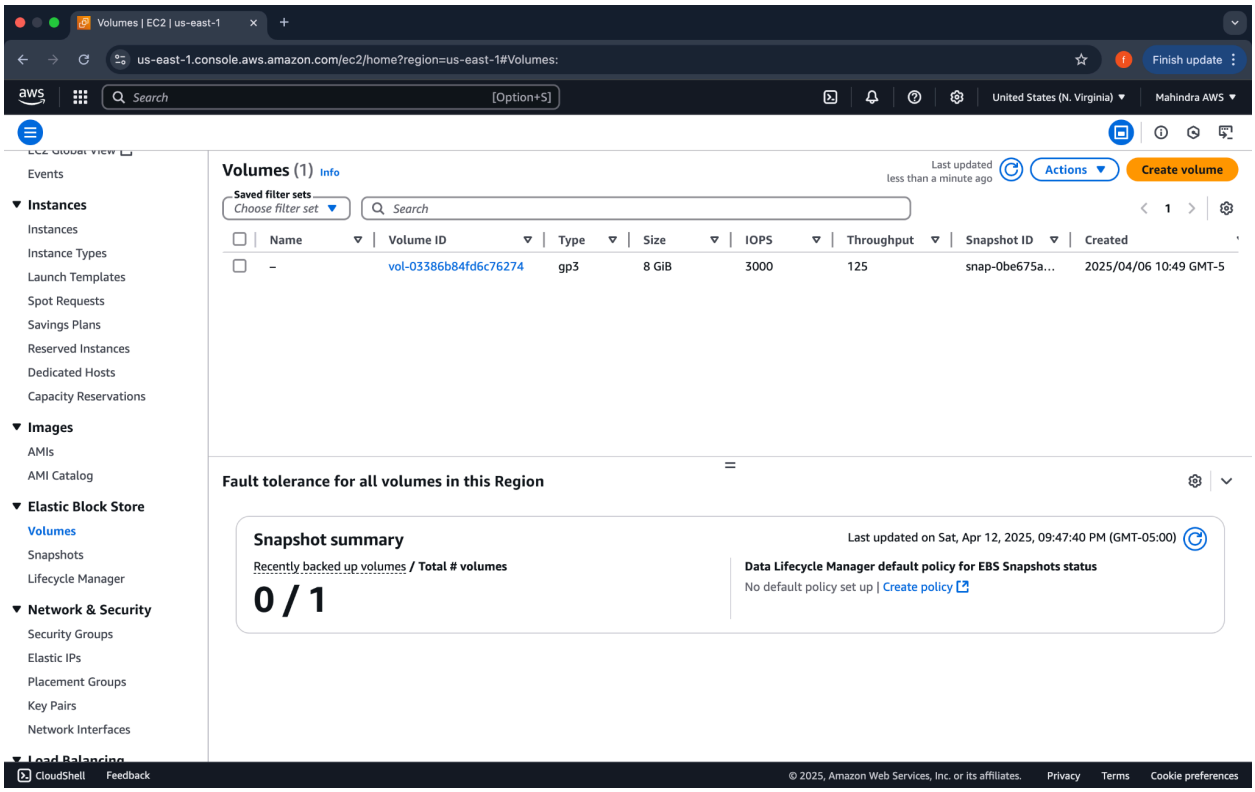
For this exercise, you'll encrypt an unencrypted volume attached to a running EC2 instance.

1. Create an EC2 instance from an unencrypted snapshot or AMI. If you already have one you can bear to part with, you may use that.

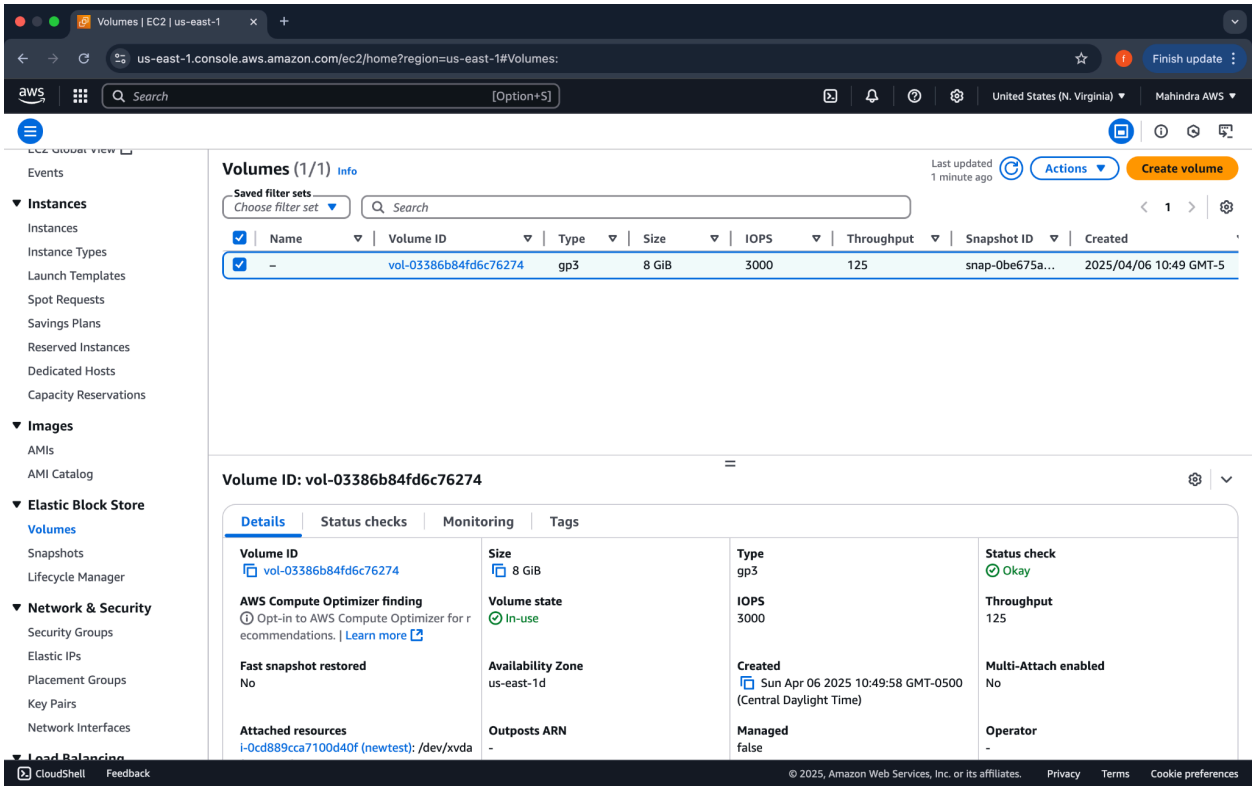

Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running
Finish update
Search [Option+S]
United States (N. Virginia)
Mahindra AWS
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Instances (1) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Pub
newtest
i-0cd889cca7100d40f
Running
t2.micro
2/2 checks passec
View alarms
us-east-1d
ec2-
Select an instance
CloudShell
Feedback
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Privacy
Terms
Cookie preferences


EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running
Finish update
Search [Option+S]
United States (N. Virginia)
Mahindra AWS
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Instances (1) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Pub
newtest
i-0cd889cca7100d40f
Running
t2.micro
2/2 checks passec
View alarms
us-east-1d
ec2-
Select an instance
CloudShell
Feedback
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Terms
Cookie preferences

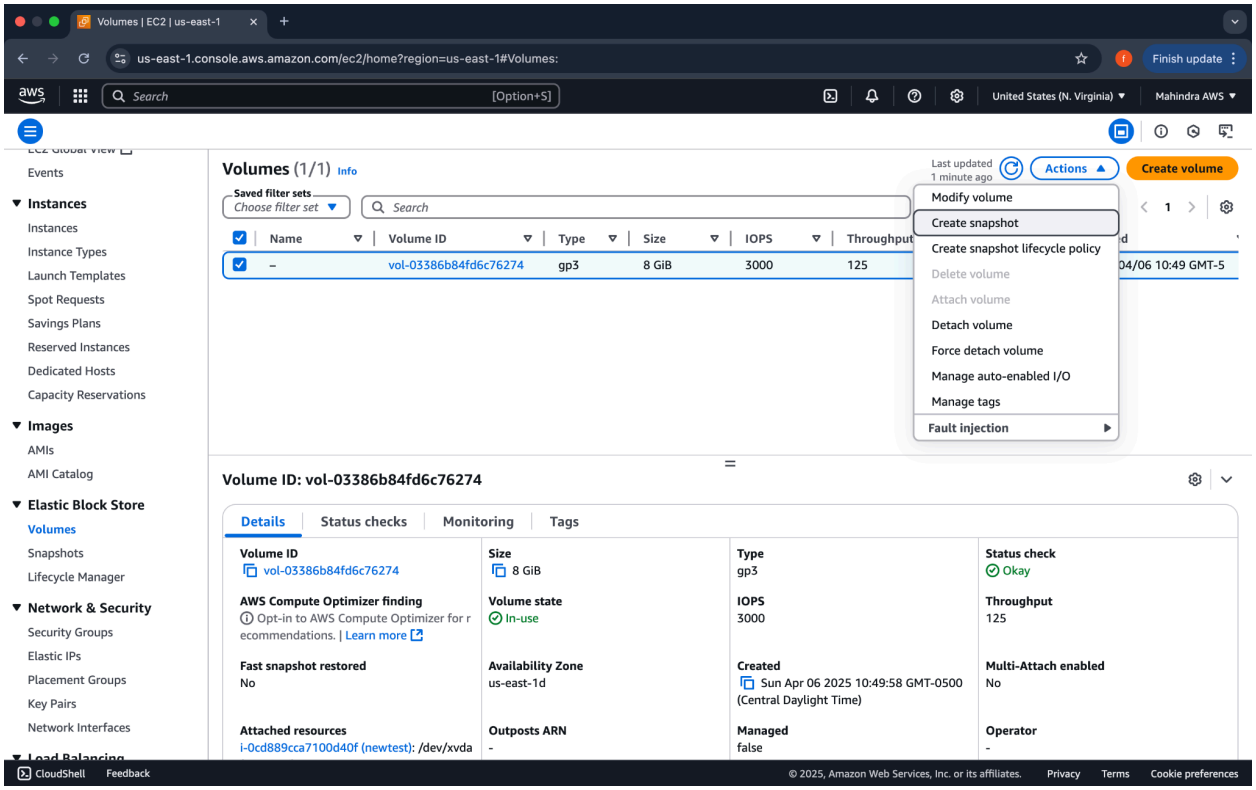
2. At the EC2 Dashboard, click Volumes in the left-side menu.

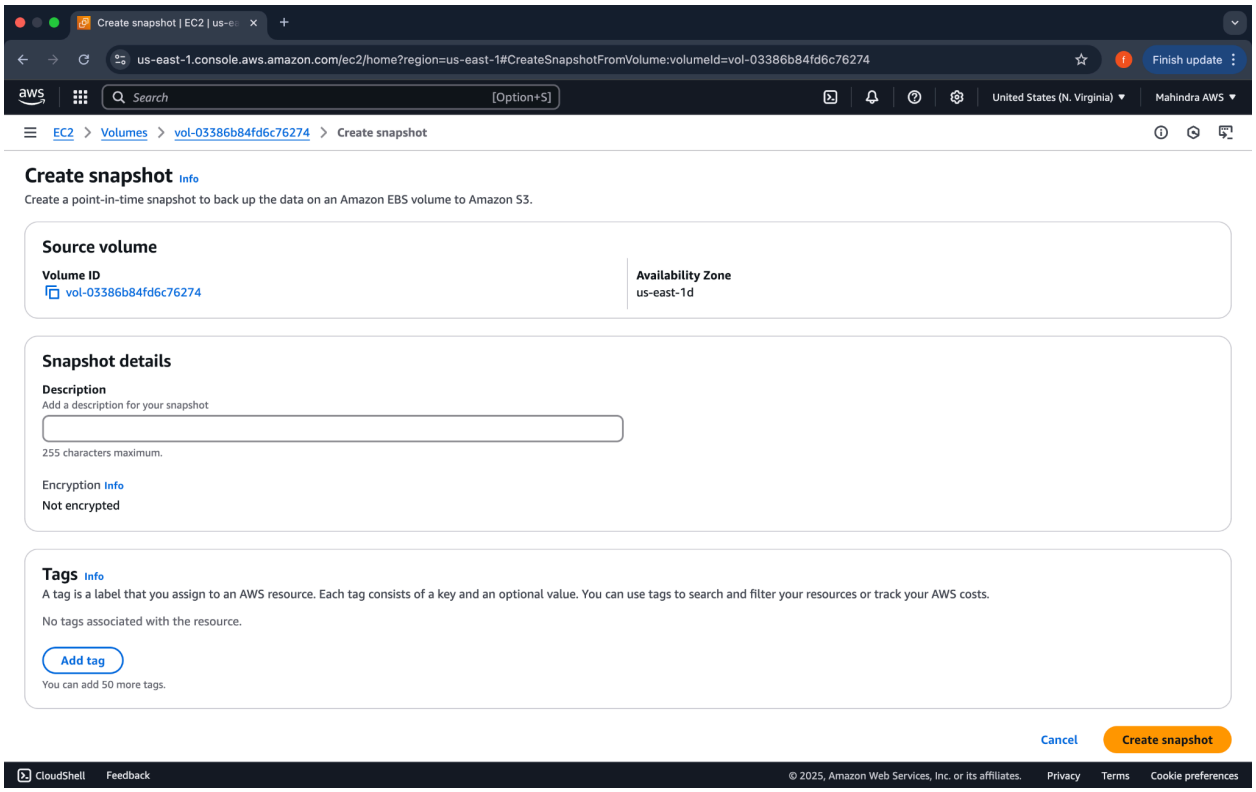


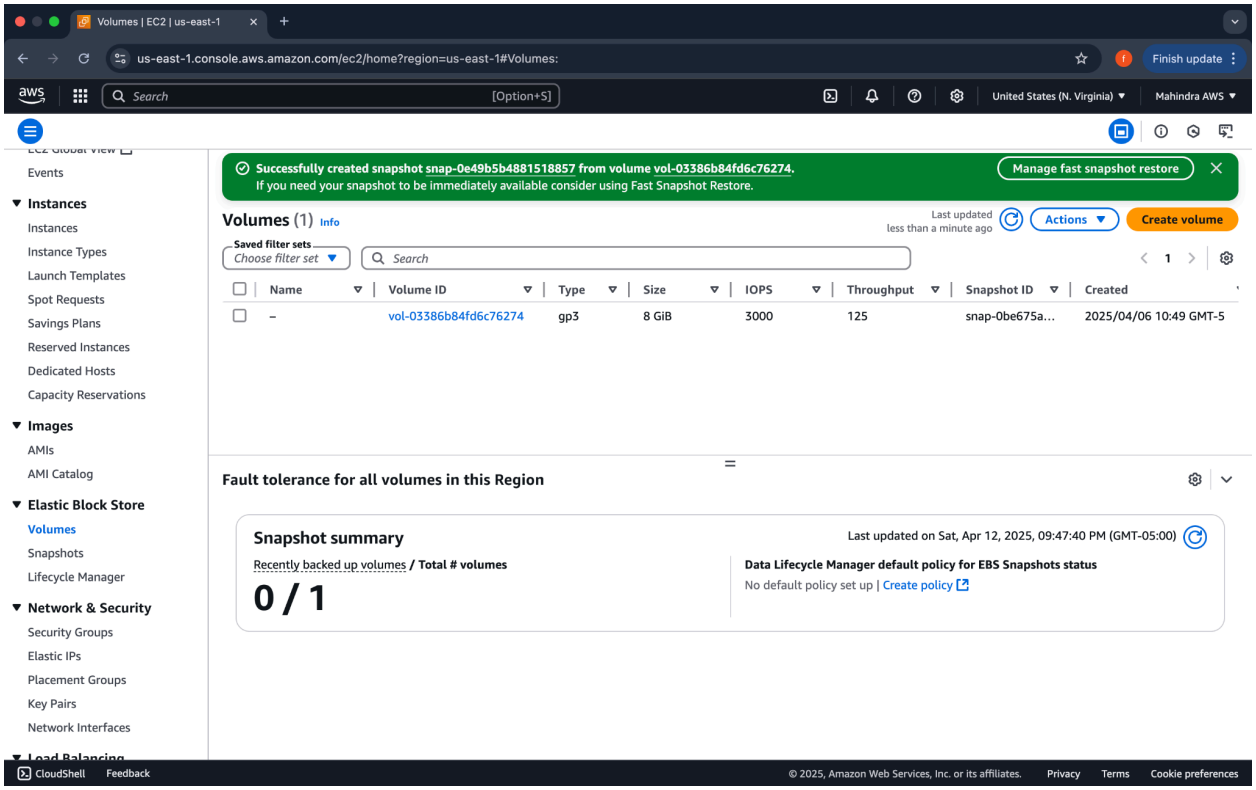
3. Select the volume attached to your instance.



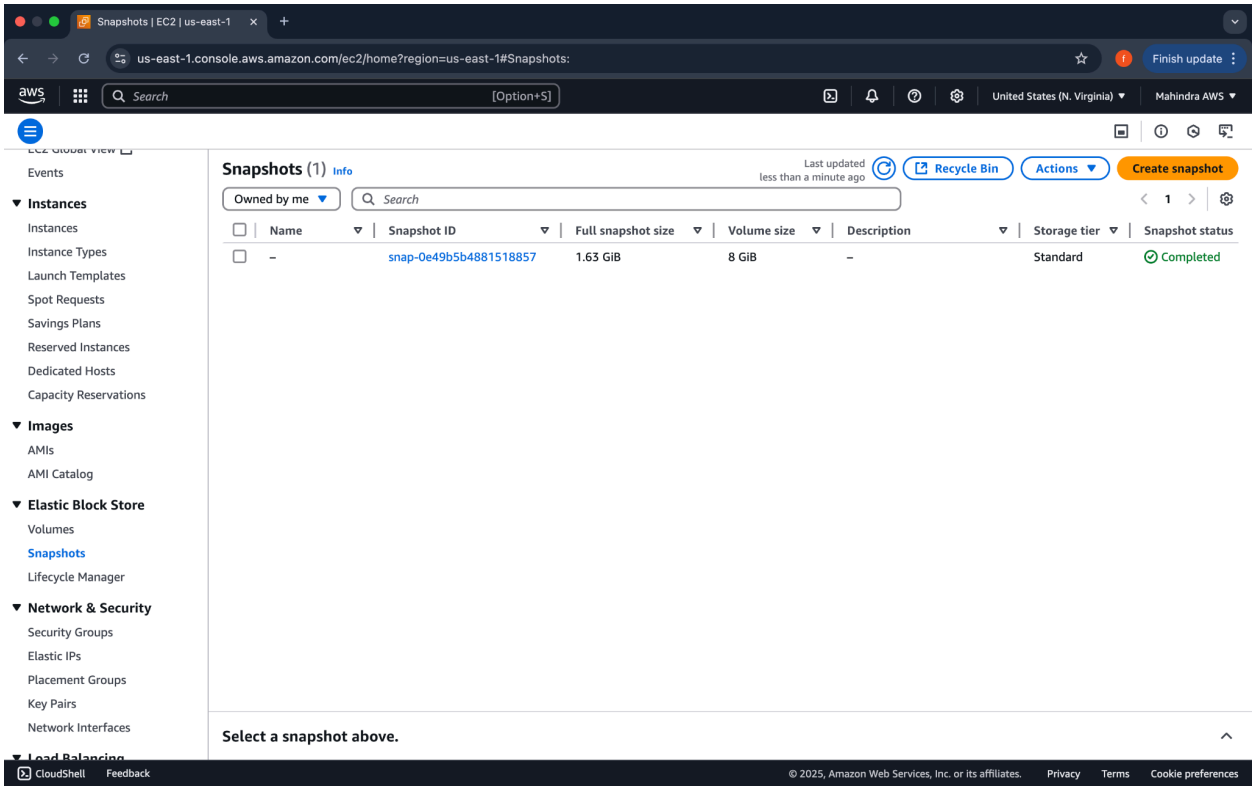
4. Under the Actions menu, click Create Snapshot. EBS will begin creating an unencrypted snapshot of the volume.


Volumes | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Volumes:
Finish update
Search
[Option+S]
United States (N. Virginia)
Mahindra AWS
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Volumes (1/1) Info
Last updated 1 minute ago
Actions
Create volume
Saved filter sets
Choose filter set
Search
Name
Volume ID
Type
Size
IOPS
Throughput
vol-03386b84fd6c76274
gp3
8 GiB
3000
125
04/06 10:49 GMT-5
Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume
Manage auto-enabled I/O
Manage tags
Fault injection
Volume ID: vol-03386b84fd6c76274
Details
Status checks
Monitoring
Tags
Volume ID
vol-03386b84fd6c76274
Size
8 GiB
Type
gp3
Status check
Okay
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
Volume state
In-use
IOPS
3000
Throughput
125
Fast snapshot restored
No
Availability Zone
us-east-1d
Created
Sun Apr 06 2025 10:49:58 GMT-0500 (Central Daylight Time)
Multi-Attach enabled
No
Attached resources
i-0cd889cca7100d40f (newtest): /dev/xvda
Outposts ARN
-
Managed
false
Operator
-
CloudShell
Feedback
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Terms
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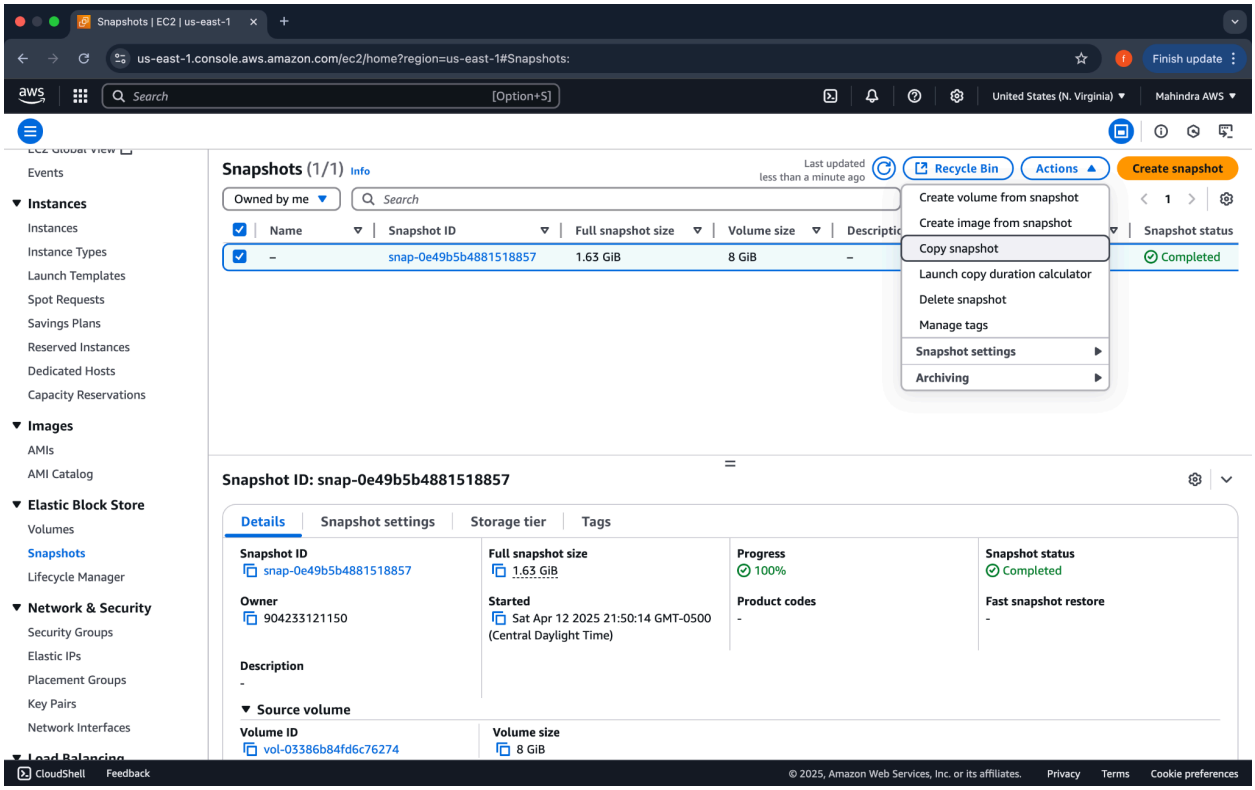

Create snapshot | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSnapshotFromVolume:volumeId=vol-03386b84fd6c76274
Finish update
Search
[Option+S]
United States (N. Virginia)
Mahindra AWS
EC2 > Volumes > vol-03386b84fd6c76274 > Create snapshot
Create snapshot Info
Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.
Source volume
Volume ID
vol-03386b84fd6c76274
Availability Zone
us-east-1d
Snapshot details
Description
Add a description for your snapshot
255 characters maximum.
Encryption Info
Not encrypted
Tags Info
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
No tags associated with the resource.
Add tag
You can add 50 more tags.
Cancel
Create snapshot
CloudShell
Feedback
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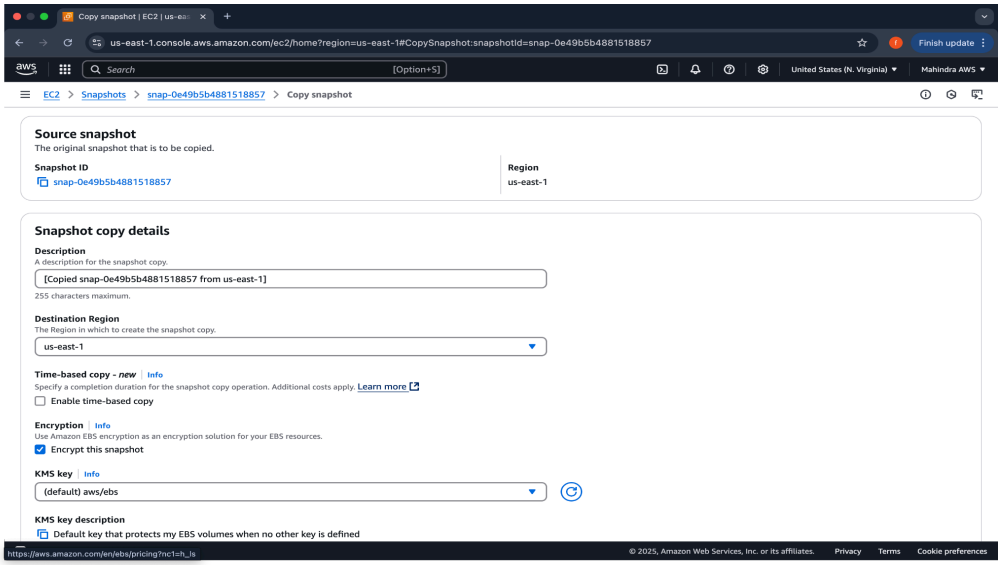
5. On the left-side menu, click Snapshots. Wait for the snapshot to complete.



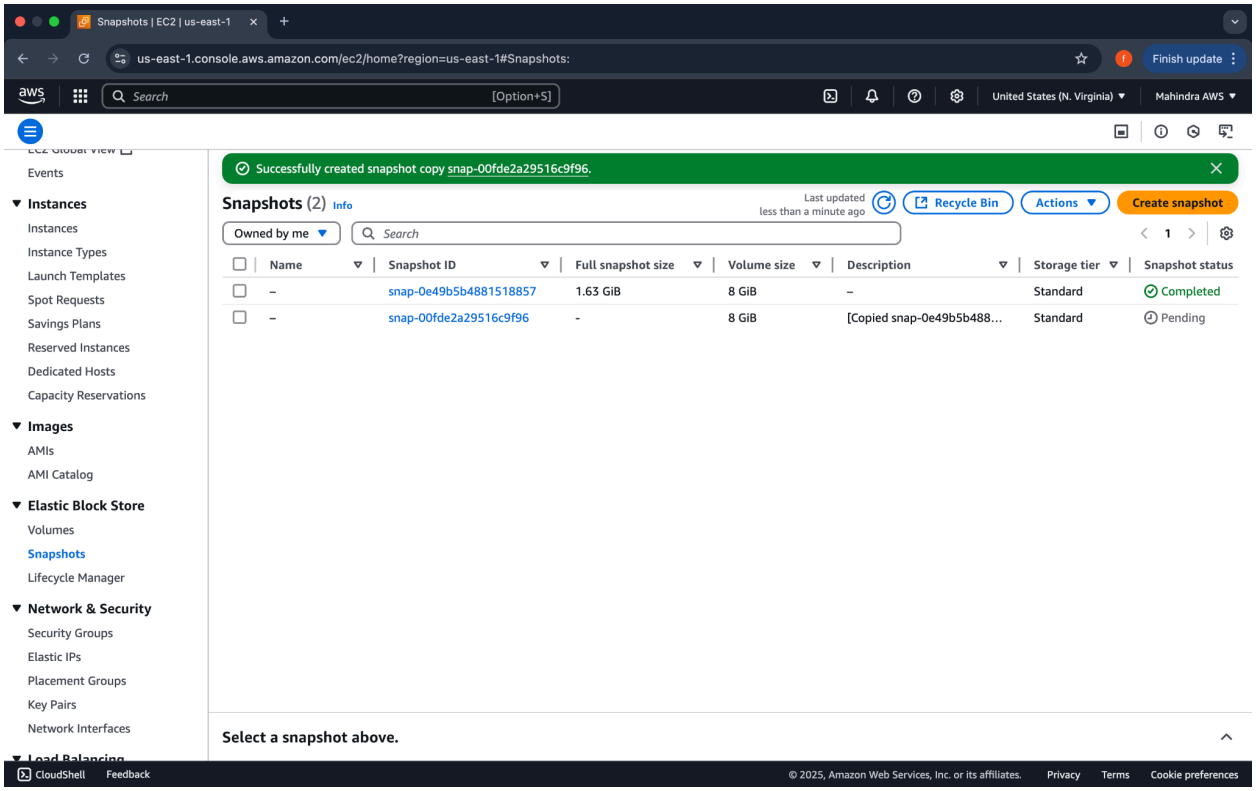
6. Select the snapshot and under the Actions menu, click Copy.



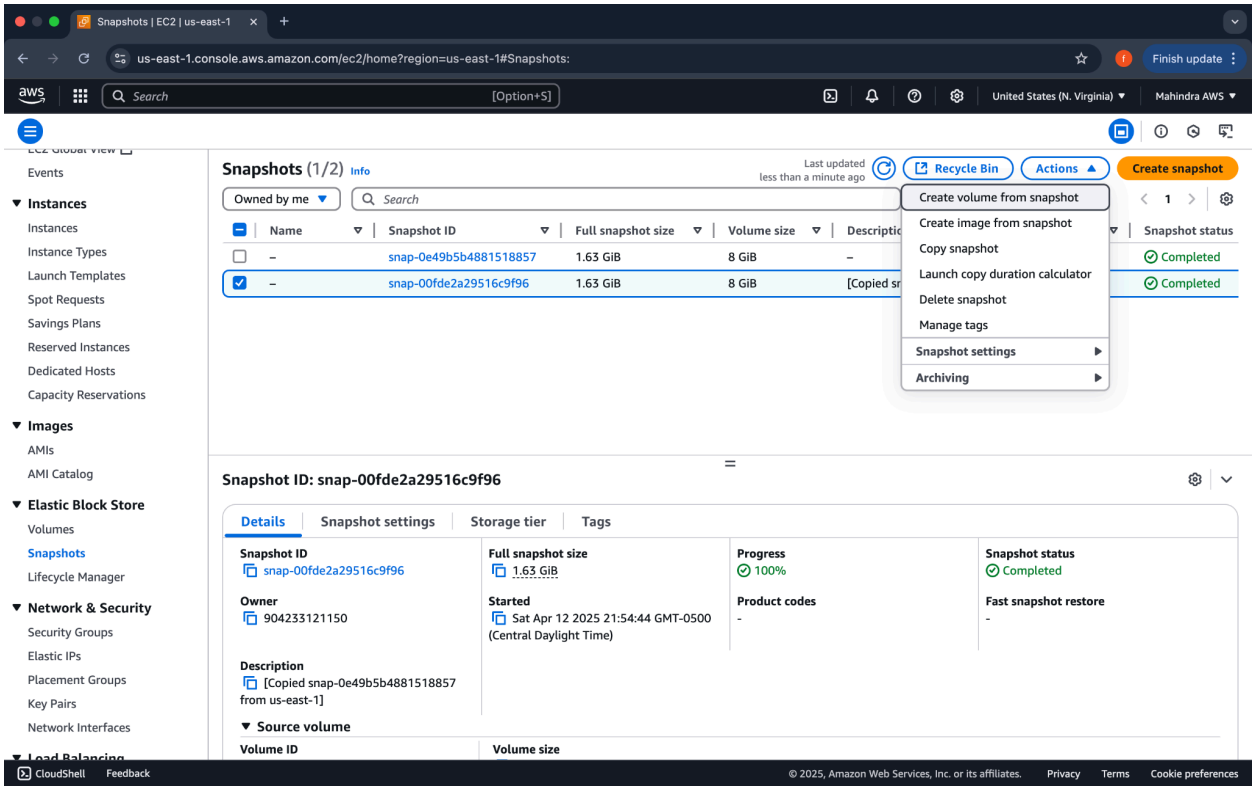
7. Select the Encrypt This Snapshot check box.



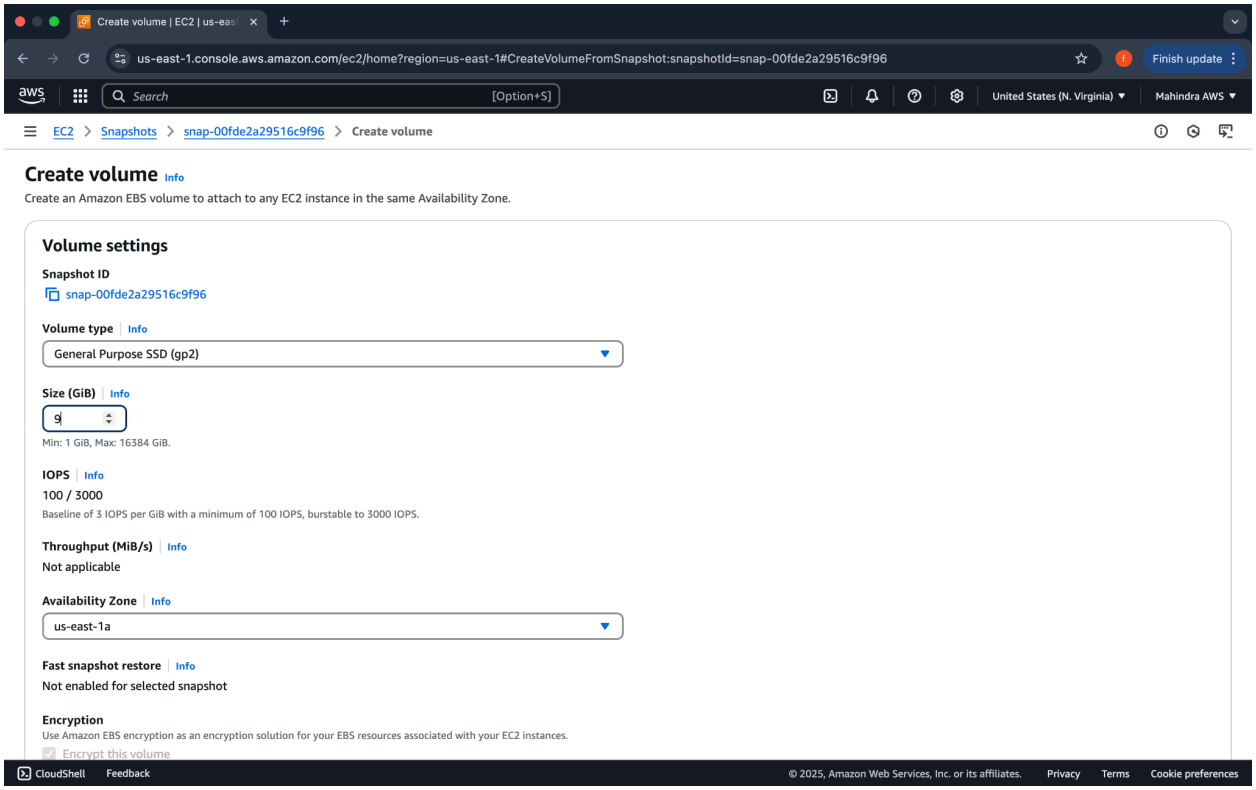
8. Next to the Master Key drop-down, select the KMS key you'd like to use to encrypt the snapshot. You can use your own customer master key or the default AWS/EBS key.
9. Click the Copy button. EBS will begin creating an encrypted copy of the snapshot. The larger the volume, the longer the encryption process takes. A good rule of thumb is about 3 gigabytes per minute.



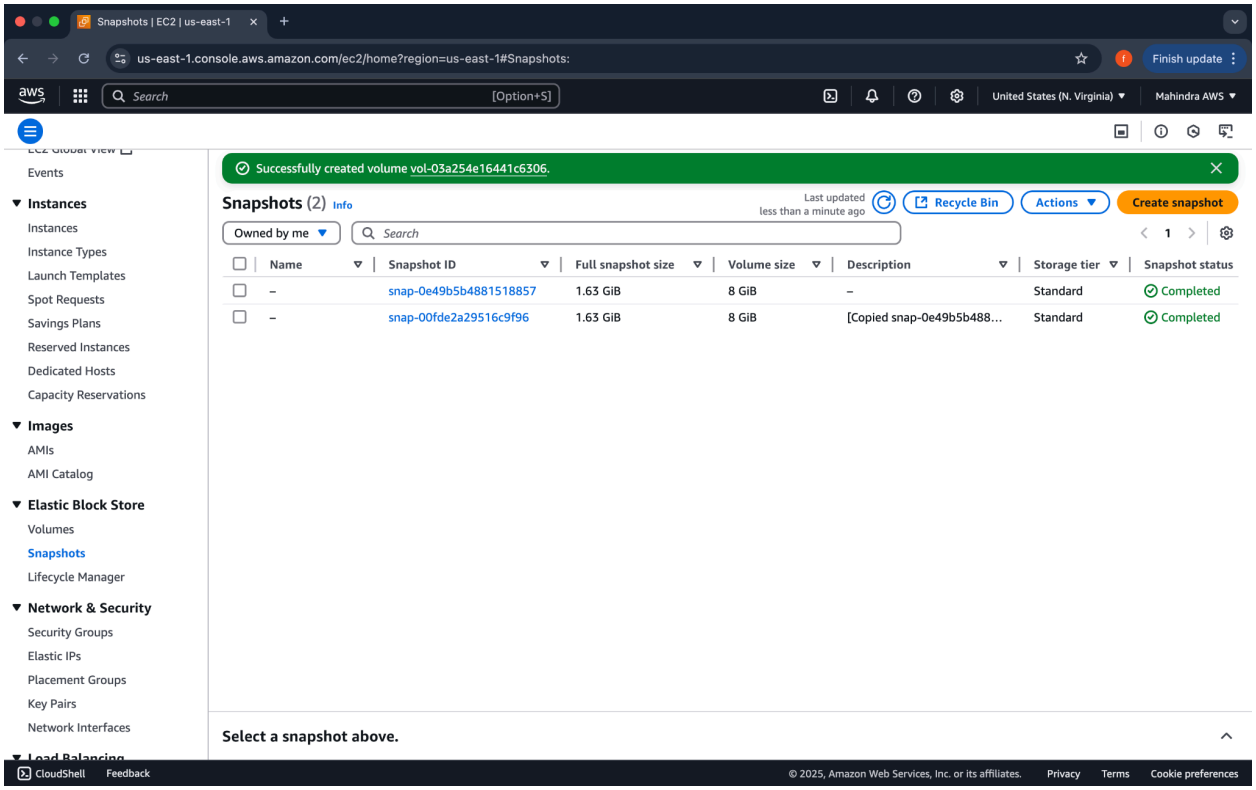
10. Select the encrypted snapshot and under the Actions menu, click Create Volume.



11. Choose the volume type, size, and availability zone, which can be different than the original volume.



12. Click the Create Volume button. The new volume will be encrypted using the same key that was used to encrypt the snapshot.


Snapshots | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Snapshots:
Finish update
Search
[Option+S]
United States (N. Virginia)
Mahindra AWS
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Successfully created volume vol-03a254e16441c6306.
Snapshots (2) Info
Last updated
less than a minute ago
Recycle Bin
Actions
Create snapshot
Owned by me
Search
1
Name
Snapshot ID
Full snapshot size
Volume size
Description
Storage tier
Snapshot status
– snap-0e49b5b4881518857 1.63 GiB 8 GiB – Standard Completed
– snap-00fde2a29516c9f96 1.63 GiB 8 GiB [Copied snap-0e49b5b488… Standard Completed
Select a snapshot above.
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